

Highlights

TreeMMM: Tree-Based Panel Marketing-Mix Attribution and Cap-Bounded Budget Action Planning

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- Open-source panel MMM connects channel attribution, automatic interaction discovery, mROI ranking, and cap-bounded HCP-period budget landing from one fitted tree response surface.
- TreeMMM achieves $17.9\% \pm 0.2\%$ non-linear attribution-share MAPE ($N=5$ seeds), improving over GLMM-Naive by $4.3 \text{ pp} \pm 0.4 \text{ pp}$ without pre-specifying interactions or distributional families.
- Automatic interaction discovery recovers 5 of 6 planted channel synergies (recall 0.83, F1 0.56), providing the mechanism behind TreeMMM's attribution and mROI advantage.
- mROI recommendations rank channels correctly (Spearman $\rho = 0.96$) and identify the optimal up/down direction for 94% of channels against the known data-generating process.
- A committed +25% rep-visit increase demonstrates the action layer: cap-bounded allocation spreads the increment across HCP-period cells, predicting +27.8% lift against a +20.7% DGP-reference response.

TreeMMM: Tree-Based Panel Marketing-Mix Attribution and Cap-Bounded Budget Action Planning

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Abstract

Marketing Mix Modeling (MMM) often stops at channel attribution or aggregate response curves, leaving analysts to translate model output into action plans by hand. We introduce TreeMMM, an open-source panel MMM package that uses gradient-boosted trees (LightGBM/XGBoost/CatBoost) [1, 2, 3] and SHAP attribution to connect four decision layers from the same fitted response surface: channel-share attribution, automatic interaction discovery, mROI ranking, and cap-bounded HCP-period budget landing. SHAP TreeExplainer provides exact Shapley values that are additive on the model’s native scale, and a distribution-aware decomposer guarantees attributions sum to predicted outcomes regardless of objective function (Poisson, Tweedie, Gamma, or Gaussian).

We benchmark TreeMMM against five baselines on four synthetic datasets with known reference decomposition (3,000 entities $\times$ 36 periods, $N=5$ seeds). Four operate on the full panel: a naive GLMM, an oracle GLMM, and a customer-level hierarchical Bayesian model in two variants (PyMC-Hier-Naive / PyMC-Hier-Oracle). One operates on aggregated

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36-row time-series: PyMC-Marketing. The headline metric is attribution-share MAPE, not predictive MAPE on the outcome variable. TreeMMM achieves $17.9\% \pm 0.2\%$ non-linear average attribution-share MAPE versus $22.2\% \pm 0.3\%$ for GLMM-Naive (gap 4.3 pp ± 0.4 pp), matching or beating the Oracle baselines (GLMM-Oracle $19.7\% \pm 0.4\%$, PyMC-Hier-Oracle $18.1\% \pm 0.3\%$) without receiving any interaction structure in advance. The mechanism is interaction discovery: TreeMMM recovers 5 of 6 planted channel synergies (precision 0.42, recall 0.83, F1 0.56; 10.9% false-positive rate), while main-effects baselines miss them unless the interaction graph is manually supplied.

A distributional GLM ablation (GLMMDist-Naive; Poisson, Tweedie, Gamma likelihoods) confirms the advantage is driven by interaction discovery and non-linear response learning, not by a distributional-family mismatch. PyMC-Marketing’s $75.1\% \pm 5.5\%$ MAPE reflects the cost of aggregating 3,000 customers to 36 time-series rows, not a limitation of Bayesian methodology: the panel-level PyMC-Hier-Naive lands within 0.5 pp of GLMM-Naive on the non-linear-DGP average.

Beyond attribution, TreeMMM’s mROI simulator recovers channel ranking and direction reliably (Spearman $\rho = 0.96$, 94% direction accuracy), then uses the same response surface to show where committed incremental budget would land. On the pharma DGP, a committed +25% rep-visit increase under a 95th-percentile per-HCP cap spreads the increment across HCP-period cells, predicting +27.8% lift against a +20.7% DGP-reference response. Absolute lift magnitude remains a synthetic model counterfactual and should be calibrated against lift studies, geo-experiments, or in-market tests before

use as a forecast.

All results are from synthetic benchmarks. Source code and demonstration scripts are available under the MIT license in the Python and R GitHub repositories cited in Code availability.

Keywords: Marketing mix modeling, SHAP attribution, gradient-boosted trees, LightGBM, interaction discovery, budget action planning, marketing analytics

1. Introduction

1.1. The Market Mix Modeling Problem

Market Mix Modeling (MMM) seeks to attribute observed commercial outcomes (sales, prescriptions, conversions) to promotional levers (advertising, sales force, digital marketing, trade promotions) while controlling for non-promotional factors (seasonality, competition, macroeconomics). The stakes are high: MMM outputs directly drive multi-million-dollar budget allocation decisions across the marketing portfolio [4].

1.2. Existing Tools

The open-source MMM ecosystem is dominated by Bayesian and ridge regression approaches [5, 6] (Table 1).

Recent academic work has explored neural architectures for MMM: NNN [7] uses transformers; CausalMMM [8] learns causal channel structure with temporal response constraints; DeepCausalMMM [9] combines GRUs with learned DAG structure, Hill saturation curves, and time-varying coefficients. None provide the same package-level combination of tree-ensemble panel

Table 1: Open-source MMM tools and their primary limitations.

Tool	Method	Key Limitation
Robyn (Meta)	Ridge regression + Nevergrad	No posteriors; Gaussian only
Meridian (Google)	Hierarchical Bayesian (MCMC)	Parametric functional forms
PyMC-Marketing	Bayesian via NUTS	Manual saturation/adstock spec
Orbit (Uber)	Bayesian structural time-series	Not purpose-built for MMM

attribution, SHAP decomposition, interaction discovery, synthetic channel-reference benchmarking, and HCP-period budget landing used here. An exploratory comparison with DeepCausalMMM is reported in [Appendix A](#).

1.3. The Gap and TreeMMM’s Contribution

To our knowledge, no open-source tree-ensemble MMM implementation combines panel-scale SHAP attribution, link-function-aware decomposition, automatic interaction discovery, synthetic channel-reference benchmarks, and cap-bounded HCP-period budget landing. The closest verifiable open-source comparator is Robyn [10], a widely adopted aggregate time-series MMM package; it does not target customer-level tree-ensemble attribution or HCP-period budget landing.

TreeMMM addresses four specific limitations shared by existing tools:

1. **Interaction discovery:** Existing MMM tools require manually specifying interaction terms. Trees discover interactions automatically through split structure.
2. **Distribution-aware modeling:** Robyn and Meridian default to Gaussian loss. TreeMMM auto-detects the outcome distribution

(Poisson, Tweedie, Gamma, Gaussian) and selects the appropriate objective function.

3. **Link-function-aware attribution:** SHAP TreeExplainer [11] computes values in margin space. For log-link models, naïve exponentiation of SHAP values violates additivity. TreeMMM’s decomposer handles this correctly.
4. **Actionable budget landing:** Aggregate MMM can rank channels but usually leaves the analyst to decide where an incremental budget actually lands. TreeMMM converts a committed channel increase into a cap-bounded HCP-period allocation using the same fitted response surface.

1.4. The “Bayesian MMM is superior” claim is regime-conditional

The conventional framing that Bayesian methods are the right default for marketing-mix modeling was forged in a specific data regime: aggregate weekly time-series with around 150 observations, ten or more collinear channels, and stakeholder demand for posterior intervals. In that regime the framing is correct. Frequentist OLS struggles with weakly identified adstock and saturation parameters [12, 13], priors regularize estimation and allow transfer of prior experimental knowledge, and hierarchical structure benefits from partial pooling. These advantages are real because of the data scarcity and aggregation that defines the classical MMM problem.

Modern pharmaceutical and B2B promotional analytics rarely produce data in that regime. The benchmark panel in this paper (3,000 customers $\times$ 36 months = 108,000 observations) sits in a different statistical environment. Four panel-data shifts move the right answer away from “default to Bayesian”:

1. **Cross-sectional variation breaks multicollinearity.** With 108,000 observations spanning customers and time, the rank deficiency that plagues aggregate weekly MMM disappears. The dominant identification path is now within-stratum cross-sectional contrast, not the national-trend separation Bayesian priors are designed to regularize.
2. **Business constraints keep counterfactuals closer to observed support.** Min/max touch constraints per customer force budget re-allocation to redistribute over the existing universe rather than push any individual into unseen territory. Marginal caps are not a proof of full multivariate support, so coverage diagnostics still matter, but the planning query is much closer to the training distribution than an aggregate saturation curve that extrapolates along a parametric form.
3. **Composite potential variables absorb observable selection bias.** A well-constructed potential composite (historical prescribing, panel size, specialty, payer mix, market-share trends) means that within strata of potential, variation in promotional intensity can be interpreted with stronger controls than an aggregate time-series model has available. This is not DML without orthogonalization, and it does not solve selection on unobservables. That residual concern is one Bayesian MMM does not solve either.
4. **Out-of-sample validation becomes more informative.** With 108,000 panel observations, time-respecting validation can test prediction shape and calibration across customer-period rows. Aggregate Bayesian MMM with 36 monthly observations has little holdout surface for validating response-curve shape.

Under these four shifts, tree-based methods with SHAP attribution are competitive with or superior to Bayesian MMM for most promotional decisions. Under the classical aggregate regime, they are not. The benchmark in Section 3 makes the empirical version of this claim concrete: panel-Bayesian and panel-frequentist baselines land within 0.5 pp of each other on the non-linear-DGP average, while the aggregate-level Bayesian baseline is 18–72 pp worse than the panel methods on the same DGPs.

1.5. What TreeMMM is and is not designed for

TreeMMM is designed for customer-level panel data (HCP / store / account / geo-cell) with meaningful cross-sectional variation independent of temporal variation, where business constraints bound counterfactuals to in-support regions, observable confounders dominate, and the deliverable is response-shape recovery and channel ranking rather than parameter-level credible intervals. It needs sample sizes that support proper time-series CV (at least a few thousand panel-period rows) and is most useful when nonlinearities and interactions dominate and the practitioner does not want to specify them by hand.

TreeMMM is also designed to bridge attribution into constrained action planning. The same fitted response surface supports three decision layers: (1) channel ranking and up/down direction, (2) budget-neutral cross-tactic reallocation under the current total budget, and (3) cap-bounded HCP-period landing for a committed increase to one tactic or to a stated tactic list. It does *not* yet solve the larger combinatorial planning problem of choosing both net-new tactic mix and HCP-period landing jointly, nor does it generate territory-level call plans with routing, sequencing, rep capacity, and

channel-cadence constraints. Those plan-level optimizers are future work.

Table 2: Decision contract for TreeMMM action planning.

Supported from the current package	Requires additional validation or future optimization
Channel attribution and rank ordering	Causal ROI claims without randomized or quasi-experimental support
Interaction hypotheses for domain review	Absolute lift forecasts without lift-study calibration
Directional mROI up/down recommendations	Launch decisions for brand-new channels with no historical variation
Budget-neutral cross-tactic reallocation	Net-new cross-tactic increment optimization
Cap-bounded HCP-period landing for a committed increase	Full field plan optimization with territory, cadence, and routing constraints

Several regimes call for Bayesian or simpler regression methods rather than TreeMMM:

- **Brand-new channels with no historical variation** — trees cannot estimate the effect of an input never seen at any non-zero level. Bayesian models with informative priors from external lift studies can. TreeMMM does not solve cold-start.

- **Decisions requiring parameter-level credible intervals** — TreeMMM provides prediction intervals via conformalized quantile regression and bootstrap resampling, not parametric posteriors. The panel-Bayesian baseline (Appendix [Appendix B.6](#)) is a drop-in replacement at the same panel scale with similar attribution accuracy.
- **Selection on unobservables** — neither paradigm solves this. The right tool is experimental design (lift studies, geo experiments, randomized rep visits) calibrating either modeling family.
- **Hierarchical sparse-cell estimation** — when many small cells hold fewer than five observations each, partial pooling via Bayesian hierarchical models dominates; trees fit each stratum independently and overfit on small cells.

1.6. *Decision branches the paper sits on*

The headline claims live on a concrete branch of the decision tree. Table [3](#) summarizes the regimes that favor each paradigm.

The four DGPs in the paper (pharma_brand, cpg_brand, saas_brand, linear_baseline) are designed to live on the right-hand branch. The linear DGP is the *honesty test*: when the data-generating process is linear and Gaussian (the natural home turf of a GLMM or Bayesian regression), TreeMMM should not dominate. The benchmark confirms this — on the linear DGP TreeMMM posts $0.4\% \pm 0.1\%$ attribution-share MAPE (multi-seed) against GLMM-Naive’s $0.3\% \pm 0.0\%$.

Table 3: Regime indicators by paradigm.

Bayesian-favored regime	TreeMMM-favored regime
Obs-to-parameter ratio < 20	Panel structure with cross-sectional variation independent of temporal
Design-matrix condition number > 30	Treatment overlap reasonable across covariate strata
Treatment variation primarily temporal, channels co-moving ($\rho > 0.7$)	Counterfactuals stay within empirical support
Credible informative priors from external evidence	Confounders largely observable
Decisions require probabilistic statements	Non-linearities and interactions dominate the response surface
Hierarchical units with sparse cells	Sample size supports proper CV

1.7. Risks of misuse, symmetric across paradigms

The paper does not argue that tree-based methods are categorically safer than Bayesian methods. They have different failure modes.

Bayesian-specific failure modes include tight priors centered on desired conclusions, identifiability problems masked by smooth posteriors, MCMC convergence theater ($\hat{R}$ at 1.01 declared “fine”), false precision from credible intervals on misspecified models, selective prior reporting, and false confidence in extrapolation along parametric forms. The prior-sigma sweep in Section 3.9 is the symmetric counterpart of seed/hyperparameter perturbations recommended for the tree side; at $n = 3,000 \times 36$ the panel-Bayesian model is insensitive to prior choice, but at smaller n this same sweep is what

would surface miscalibration.

Tree-specific failure modes include confident-looking SHAP attributions in collinear settings without sensitivity checks, flat extrapolation outside support hidden by smooth predictions, weak parametric uncertainty quantification, and the temptation to treat prediction accuracy as causal validity. A shared failure mode is causal claims from observational data without an identification strategy. Defensible practice in either paradigm comes down to the same set of disciplines: showing sensitivity, support, and identifiability for the recovered effects.

1.8. Diagnostics worth running

Five diagnostics determine whether either paradigm is in defensible territory on a given dataset. All are exposed via the package’s diagnostics API (`treemmm.core.diagnostics`).

- **Coverage check** — count training observations within a neighborhood of each proposed counterfactual input; under ~ 30 nearest neighbors signals extrapolation regardless of method.
- **Identifiability check** — refit with prior variance halved/doubled (Bayesian) or seed/hyperparameter perturbations (tree). The half-and-double-sigma version is implemented for the panel-Bayesian baseline and runs automatically (see Section 3.9).
- **Treatment-overlap check** — propensity scores for promotional intensity; common support below 80% across covariate strata means no method recovers causal effects without strong assumptions.

- **Variation decomposition** — share of total predictor variance living within-unit (temporal) vs. between-unit (cross-sectional).
- **Effective sample size per parameter** — `arviz.ess` / parameter count (Bayesian), training rows / leaves at max depth (tree). Below ~ 20 obs/parameter, both paradigms are weakly identified.

1.9. Scope: Synthetic-Only Benchmarks

All benchmarks reported in this paper are on synthetic data-generating processes calibrated against pharma, CPG, and SaaS prior distributions (Section 2.1). Real-world validation against held-out outcomes, lift-study calibration, or refit stability on a real panel is deferred to follow-up work. The contribution is methodological: a panel-MMM tree-based building block (TreeMMM) with documented behavior on canonical synthetic regimes, an apples-to-apples panel-Bayesian comparison that decouples paradigm from aggregation, and an open-source reproducible package. Regime-applicability guidance (Section 4.1) is anchored on the synthetic DGPs and should be revalidated when porting to a specific brand or panel.

2. Experimental Design

2.1. Synthetic Datasets

We evaluate on four synthetic datasets with known ground-truth DGPs (Table 4). All non-linear datasets use heterogeneous customer sensitivity (HCS), where each customer draws a latent sensitivity vector from a segment-specific multivariate normal distribution. Each dataset uses 3,000 entities $\times$ 36 periods.

Table 4: Synthetic dataset specifications.

Dataset	N	Distribution	Channels	Non-lin	Interactions	HCS
Pharma	$3K \times 36$	NegBin ($r=5$)	6	log, sqrt, lin	3	2 seg
CPG	$3K \times 36$	Tweedie ($p=1.5$)	5	sqrt, log, lin	1	3 seg
SaaS	$3K \times 36$	ZI-Gamma (ZI=10%)	5	sqrt, log	2	2 seg
Linear	$3K \times 36$	Gaussian	3	None	None	None

The pharma DGP functional form (illustrative) is:

$$\lambda_{i,t} = \exp \left(\begin{array}{l} \beta_0 + \sum_k \beta_{i,k} f_k(x_{i,k,t}) \\ + \gamma_1 x_{\text{rep},i,t} x_{\text{samp},i,t} + \gamma_2 x_{\text{dtc},i,t} x_{\text{rep},i,t} \\ + \gamma_3 x_{\text{peer},i,t} x_{\text{rep},i,t} + \delta Z_{i,t} \end{array} \right), \quad (1)$$

$$y_{i,t} \sim \text{NegBin}(\lambda_{i,t}, r=5).$$

where f_k is a channel-specific response function (log, sqrt, or linear) and $\beta_{i,k}$ is drawn from a segment-specific multivariate normal (heterogeneous customer sensitivity). The pharma DGP includes targeting bias [14] (reps visit high-potential HCPs more) and channel correlation, making it the hardest dataset. The linear dataset is the honesty test: a GLMM should match or beat TreeMMM when the true relationship is linear.

2.2. Evaluation Metrics

1. **Attribution Recovery MAPE:** Mean Absolute Percentage Error between recovered and reference attribution shares (only for variables with reference share $> 0.5\%$). Reference shares are computed as the L1 norm of mean-centered DGP component contributions, normalized

to sum to 1 (a variance-attribution heuristic, not the Shapley decomposition of the DGP function — see Section 4.5.2).

2. **Rank Correlation:** Spearman ρ between recovered and true attribution rankings.
3. **Interaction Detection:** Whether both variables in a planted interaction exceed 3% SHAP importance.
4. **HCS Recovery:** Spearman ρ between true latent customer sensitivity and customer-level mean $|\phi_i|$.
5. **Distribution Matching:** Whether the correctly matched objective outperforms the mismatched objective.
6. **Predictive Accuracy:** R^2 and WMAPE on held-out test folds.

2.3. Benchmark Configuration

TreeMMM is trained with LightGBM [1] using 20 Optuna trials per fold, searching over learning rate, number of leaves, minimum child samples, L1/L2 regularization, and feature/bagging fractions. Maximum tree depth is constrained to [3, 5]; monotone constraints are enabled for all promotional variables. Attribution MAPE is computed by comparing L1-centered SHAP-derived channel shares against L1-centered ground-truth shares on the margin (log/link) scale, methodologically consistent with SHAP’s native decomposition.

3. Results

Note on seed reporting. Headline numbers are multi-seed ($N = 5$, seeds 0–4) with standard errors, reported in Table 5. Table 6 reports single-

seed exploration (seed = 42) for the distributional-GLM ablation; comparisons there should be read as indicative, not as precise multi-seed estimates. Where the body cites a percentage without a $\pm$ standard error, the multi-seed Table 5 value applies.

3.1. Attribution Recovery

Figure 1 visualizes the headline MAPE comparison across all four datasets; Tables 5 and 6 provide the underlying numerical detail. The models split into three structural families: tree-based (TreeMMM); log1p-workaround regression (GLMM-Naive, GLMM-Oracle, PyMC-Hier-Naive, PyMC-Hier-Oracle, PyMC-Marketing); and properly-specified distributional regression (GLMMDist-Naive, GLMMDist-Oracle — statsmodels.GLM with the correct exponential-family likelihood per DGP). Table 5 covers the log1p-workaround family; Table 6 adds the distributional family.

Key findings. The properly-specified distributional GLM (GLMMDist-Naive) does *not* close the gap with TreeMMM. On average across the three non-linear DGPs, GLMMDist-Naive (25.2%) is marginally *worse* than GLMM-Naive (24.0%) — gap closed = -1.2 pp (negative). TreeMMM’s lead grows from 5.7 pp over GLMM-Naive to 6.9 pp over GLMMDist-Naive. GLMMDist-Oracle with planted interactions (18.7%) nearly matches TreeMMM (18.3%), confirming: if the analyst knows both the link and the interactions, a correctly specified regression reaches TreeMMM’s accuracy. The value of TreeMMM is that it achieves this without requiring either piece of oracle knowledge.

Key observations:

Table 5: Attribution Recovery (share-MAPE, mean $\pm$ SE across $N=5$ seeds, seeds 0-4) — log1p-workaround baselines. Full-scale: 3,000 entities $\times$ 36 periods.

Dataset	TreeMM	GLMM-N	GLMM-O	PyMC-H-N	PyMC-H-O	PyMC-Mktg	Rank ρ
Pharma (NegBin)	14.5 $\pm$ 0.3	19.2 $\pm$ 0.7	25.2 $\pm$ 0.4	20.6 $\pm$ 0.5	20.5 $\pm$ 0.4	92.9 $\pm$ 11.2	1.000
CPG (Tweedie)	22.5 $\pm$ 0.3	29.0 $\pm$ 0.3	21.9 $\pm$ 1.0	29.0 $\pm$ 0.3	21.9 $\pm$ 0.9	96.1 $\pm$ 11.1	0.900
SaaS (ZI-Gamma)	16.7 $\pm$ 0.2	18.4 $\pm$ 0.5	12.0 $\pm$ 0.4	18.5 $\pm$ 0.5	12.0 $\pm$ 0.3	36.3 $\pm$ 5.2	0.900
Linear (Gaussian)	0.4 $\pm$ 0.1	0.3 $\pm$ 0.0	0.3 $\pm$ 0.0	0.1 $\pm$ 0.0	0.1 $\pm$ 0.0	21.2 $\pm$ 5.5	1.000
Non-linear avg	17.9$\pm$0.2	22.2 $\pm$ 0.3	19.7 $\pm$ 0.4	22.7 $\pm$ 0.3	18.1 $\pm$ 0.3	75.1 $\pm$ 5.5	0.933

Abbreviations: GLMM-N = GLMM-Naive; GLMM-O = GLMM-Oracle; PyMC-H-N = PyMC-Hier-Naive;

PyMC-H-O = PyMC-Hier-Oracle; PyMC-Mktg = PyMC-Marketing. Linear SE not meaningful (near-zero MAPE).

Table 6: Attribution Recovery (share-MAPE) — properly-specified distributional GLM (statsmodels.GLM with Poisson for pharma, Tweedie for CPG, Gamma for SaaS, Gaussian for linear; no random effects). Single-seed (seed = 42).

Dataset	TreeMMM	GLMMDist-N	GLMMDist-O	GLMM-N (ref)	Gap closed [†]
Pharma (NegBin)	15.6%	20.5%	26.1%	21.6%	+1.1 pp
CPG (Tweedie)	24.5%	35.5%	20.0%	32.2%	−3.3 pp
SaaS (ZI-Gamma)	14.7%	19.5%	10.1%	18.3%	−1.2 pp
Linear (Gaussian)	0.3%	0.1%	0.1%	0.1%	0.0 pp
Non-linear avg	18.3%	25.2%	18.7%	24.0%	−1.2 pp

[†] Gap closed = GLMM-Naive MAPE − GLMMDist-Naive MAPE. Positive =

proper likelihood improves; negative = proper likelihood is worse. TreeMMM lead over GLMMDist-Naive = 6.9 pp (vs. 5.7 pp over GLMM-Naive). GLMMDist-N = GLMMDist-Naive; GLMMDist-O = GLMMDist-Oracle.

- **Consistent advantage on non-linear DGPs:** TreeMMM beats all three Naive baselines on every non-linear dataset. The distributional-GLM experiment confirms the advantage is not an artifact of the log1p link workaround.
- **GLMM-Oracle provides the regression ceiling:** A perfectly specified regression will outperform a flexible learner when the analyst has complete domain knowledge. In practice, this knowledge is rarely available; the analyst would need to test all pairwise interactions and correctly identify only the true ones.
- **PyMC-Marketing illustrates the aggregation penalty:** $75.1\% \pm 5.5\%$ non-linear average MAPE ($N=5$ seeds) reflects the collapse of

3,000 customers to 36 aggregate rows. PyMC-Hier-Naive at the same panel scale lands within 0.5 pp of GLMM-Naive on the non-linear-DGP average (pharma per-DGP gap 1.44 pp), isolating the aggregation effect from the inference paradigm.

- **Linear honesty:** TreeMMM (0.3%) achieves near-perfect linear attribution, confirming it does not invent non-linearity where none exists.

3.2. Interaction Discovery

TreeMMM detected 5 of 6 planted interactions across the non-linear datasets (Tables 7 and 8; Figures 2a and 2b visualize the recovered attribution shares and the precision-recall trade-off, respectively). GLMM-Naive cannot detect interactions it was not specified to include.

The one missed interaction (`peer_programs` $\times$ `rep_visits`, strength = 0.30) has the weakest planted strength and involves `peer_programs`, which has the lowest marginal weight (0.8) among pharma channels. The false-positive rate is 10.9% of non-planted candidate pairs. False positives are interpretable co-movement signals (e.g., TV co-scheduling in CPG, customer engagement clustering in SaaS), not random noise.

Threshold sensitivity. A 5×5 grid sweep over SHAP importance $\in \{2\%, 3\%, 5\%, 7\%, 10\%\}$ and $|\rho_s| \in \{0.05, 0.08, 0.10, 0.15, 0.20\}$ (visualized in Figure 12) shows F1 ranging from 0.40 to 0.59 across 25 combinations, with a flat plateau near the default. Recall is the stable axis: any importance threshold $\leq 3\%$ delivers recall = 0.83 regardless of the correlation threshold. Three key corners are summarized in Table 9.

Table 7: Interaction detection — full confusion matrix across non-linear datasets.

Dataset	Planted Interaction	Strength	Detected	FP pairs flagged
Pharna	rep_visits $\times$ samples	0.60	Yes	—
Pharna	dtc_advertising $\times$ rep_visits	0.40	Yes	—
Pharna	peer_programs $\times$ rep_visits	0.30	No	—
Pharna	(non-planted, 25 pairs)	—	—	dtc_advertising $\times$ samples
CPG	digital_spend $\times$ trade_promo	0.35	Yes	—
CPG	(non-planted, 20 pairs)	—	—	tv_grps $\times$ digital_spend; tv_grps $\times$ trade_promo
SaaS	content_downloads $\times$ event_attendance	0.40	Yes	—
SaaS	csm_meetings $\times$ sdr_outreach	0.25	Yes	—
SaaS	(non-planted, 19 pairs)	—	—	4 pairs (sdr/content/event/csm co-movement)

Table 8: Per-dataset precision, recall, F1 for interaction detection (threshold: 3% SHAP importance, $|\rho_s| > 0.10$).

Dataset	Planted	Candidates	TP	FP	FN	Precision	Recall	F1
Pharma	3	28	2	1	1	0.67	0.67	0.67
CPG	1	21	1	2	0	0.33	1.00	0.50
SaaS	2	21	2	4	0	0.33	1.00	0.50
Aggregate	6	70	5	7	1	0.42	0.83	0.56

Table 9: Interaction detection at three threshold corners (aggregated, non-linear DGPs).

Setting	SHAP imp.	$ \rho_s $	TP	FP	FN	Precision	Recall	F1
Lax	2%	0.05	5	14	1	0.26	0.83	0.40
Default	3%	0.10	5	7	1	0.42	0.83	0.56
Strict	5%	0.15	3	2	3	0.60	0.50	0.55

3.3. Distribution Matching

Using the correct objective improves attribution recovery by 50–56% relative to the mismatched objective (Table 10; Figure 3 visualizes the effect), validating that distribution-appropriate objective selection is a meaningful modeling decision.

3.4. Heterogeneous Customer Sensitivity Recovery

Customer-level sensitivity recovery shows moderate Spearman correlations, with the strongest recovery on variables with wide heterogeneity across segments (Figure 4). On CPG, in-store display shows the highest recovery

Table 10: Distribution matching: correct vs. mismatched objective (TreeMMM).

DGP	Correct Objective	Correct MAPE	Mismatched MAPE	Rel. Improvement
Pharma (Count)	Poisson	12.1%	24.5% (Gaussian)	50.5%
Linear (Gaussian)	Gaussian	1.2%	2.7% (Poisson)	56.3%

($\rho = 0.48$), consistent with the wide HCS spread between small stores (sensitivity 1.6) and large stores (0.4). SaaS shows consistent positive correlations ($\rho = 0.08$ – 0.27); pharma recovery is weaker ($\rho < 0.14$), likely because targeting bias masks the true heterogeneous sensitivity signal.

3.5. Computation Time

In the single-seed reference timing run at benchmark scale (3,000 entities $\times$ 36 periods, 20 Optuna trials), TreeMMM takes 51–64 seconds per dataset on a 36-core CPU. GLMM-Naive takes 27–55 s; GLMM-Oracle 34–56 s. PyMC-Marketing takes 17–53 s on 36 aggregate rows. PyMC-Hier-Naive and PyMC-Hier-Oracle take 88–108 s per dataset using `nutpie` (4 chains $\times$ 1,000 tuning + 1,000 draws). An organization running TreeMMM across 50 brands completes attribution in under 1.5 hours with a single configuration file. Speed comparison is visualized in Figure 5.

3.6. Predictive Accuracy

TreeMMM achieves $R^2 > 0.5$ on all four datasets: 0.55 (pharma), 0.62 (CPG), 0.58 (SaaS), 0.95 (linear) (Figure 6). GLMM-Naive shows catas-

trophically negative R^2 on pharma ($-826,676$) because the log-transform MixedLM approach is severely misspecified for count-valued data. PyMC-Marketing shows an instructive disconnect: $R^2 = 0.50$ on pharma yet attribution MAPE = 80.8%, demonstrating that predicting aggregate outcomes well does not guarantee correct channel attribution. Predicted-vs-actual calibration plots (Figure 13) reveal the severity of GLMM-Naive’s breakdown on pharma more clearly than R^2 alone: predictions range over five orders of magnitude while actuals span less than two.

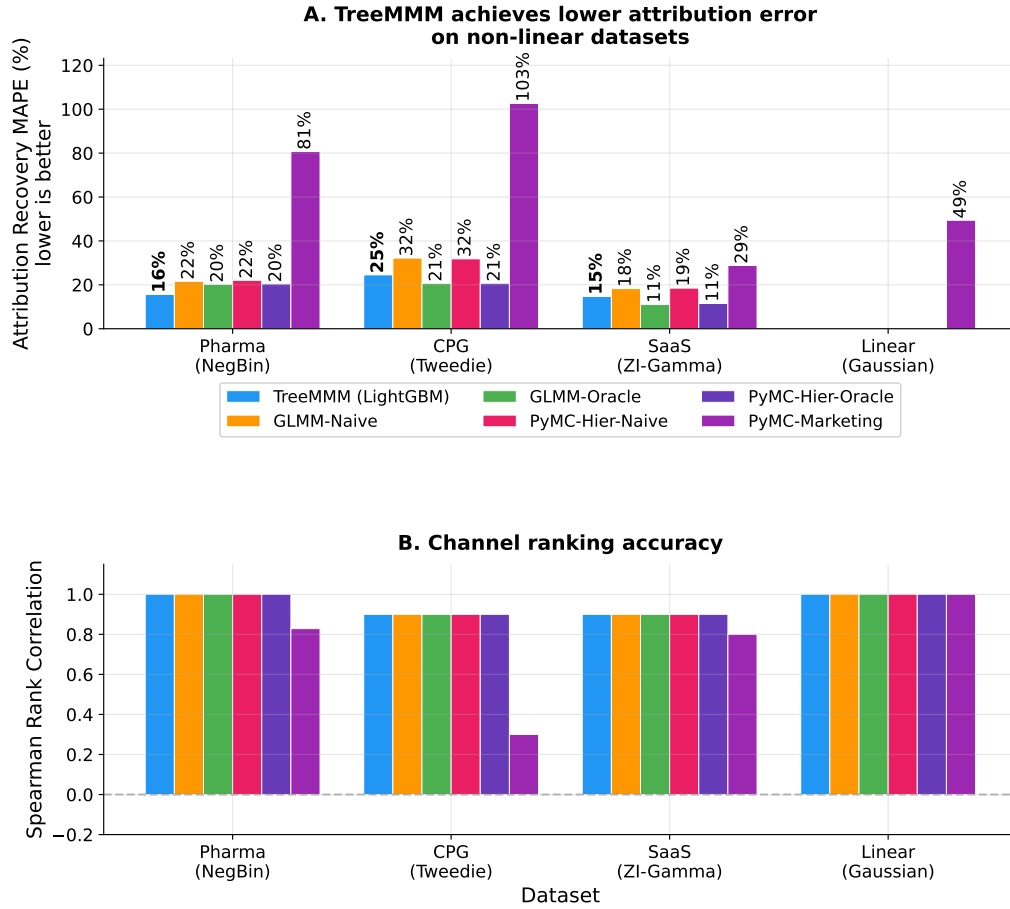


Figure 1: Attribution recovery across models and datasets. Bars show share-MAPE; lower is better. TreeMMM (orange) achieves 17.9% non-linear average MAPE vs. 22.2% for GLMM-Naive (multi-seed, $N = 5$ seeds). PyMC-Marketing (grey) incurs high error due to aggregation to 36 rows.

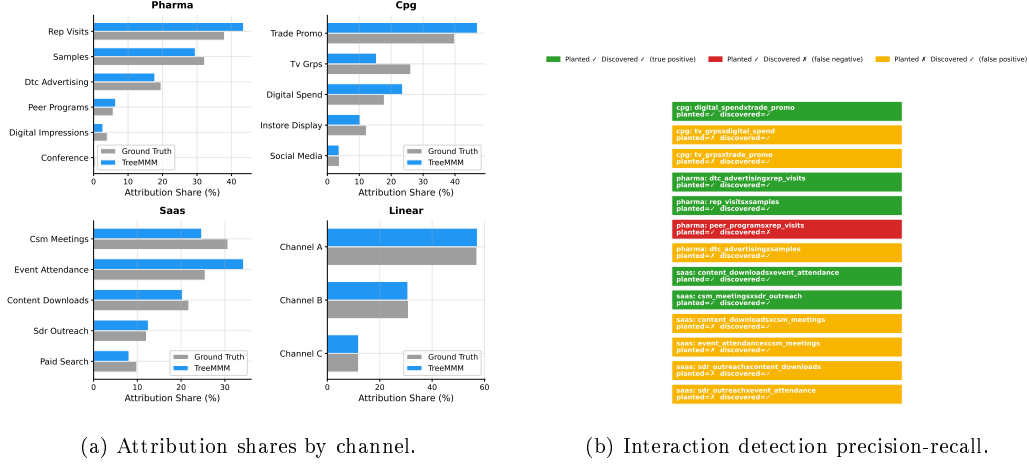


Figure 2: Left: per-channel attribution shares from TreeMMM vs. ground truth on the pharma benchmark. Right: interaction detection precision/recall trade-off at varying SHAP importance thresholds.

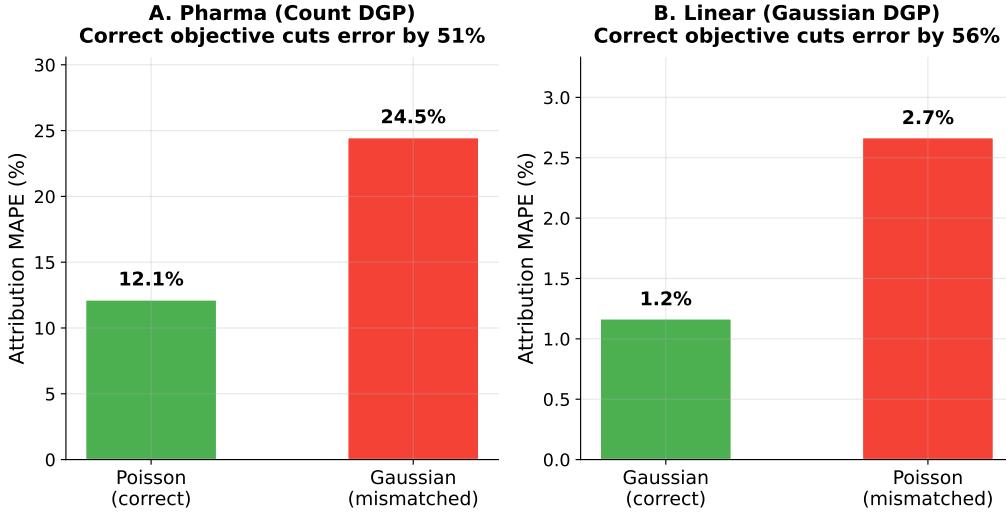


Figure 3: Distribution matching: correct vs. mismatched objective. Using the correct objective (Poisson for count outcomes, Gaussian for symmetric outcomes) improves attribution MAPE by 50–56% relative.

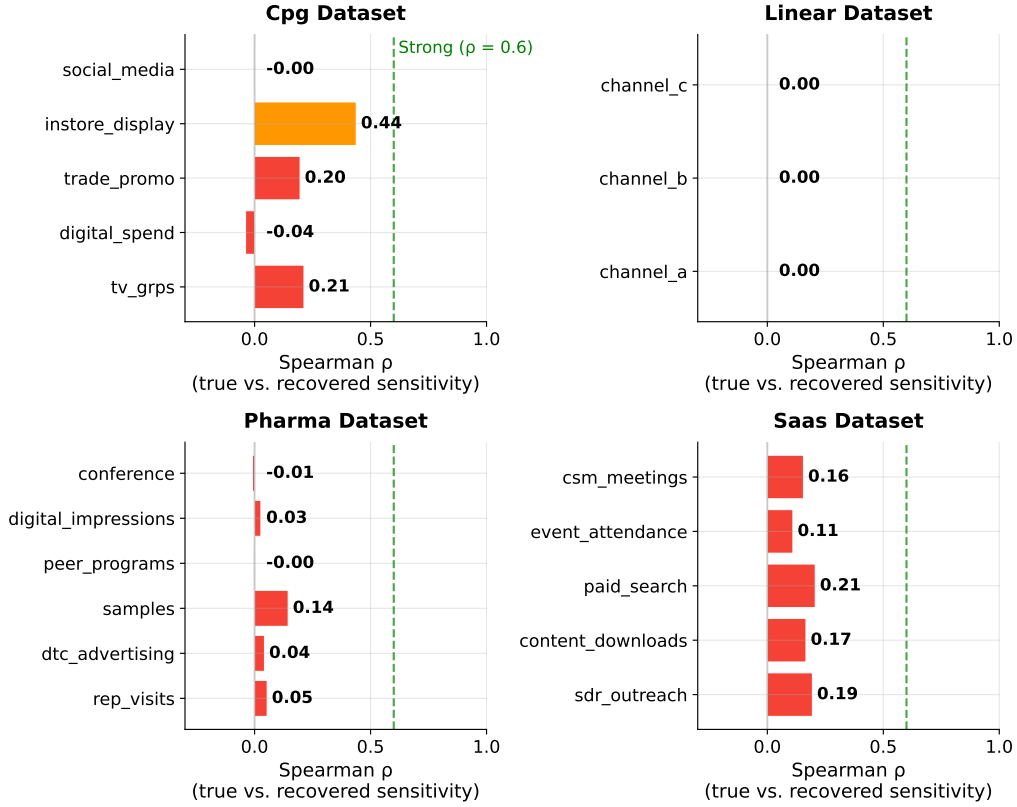


Figure 4: Heterogeneous Customer Sensitivity (HCS) recovery: Spearman ρ between true latent customer sensitivity and customer-level mean $|\phi_i|$. Strongest recovery on CPG in-store display ($\rho = 0.48$).

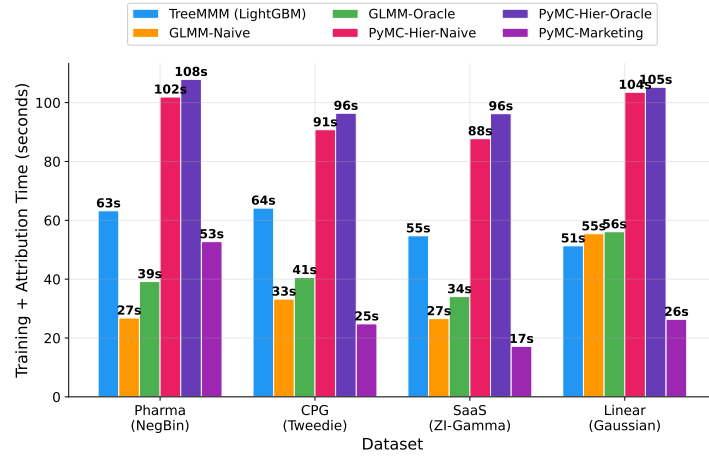


Figure 5: Computation time per dataset at $3,000 \times 36$ benchmark scale in the single-seed reference timing run. TreeMMM: 51–64 s; GLMM variants: 27–56 s; PyMC-Hier: 88–108 s; PyMC-Marketing: 17–53 s (on 36 aggregate rows).

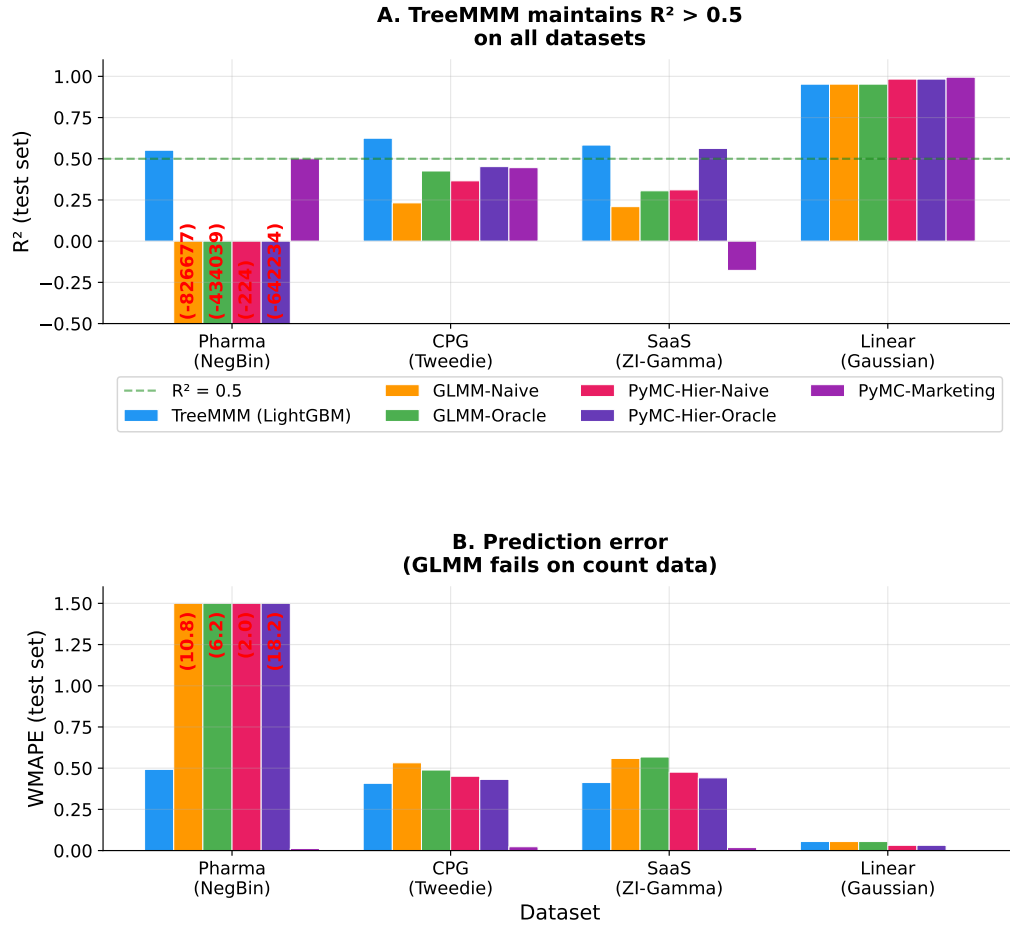


Figure 6: Predictive accuracy (R^2) by model and dataset. TreeMMM achieves $R^2 > 0.5$ on all datasets. GLMM-Naive shows catastrophically negative R^2 on pharma ($-826,676$) due to log-transform MixedLM misspecification.

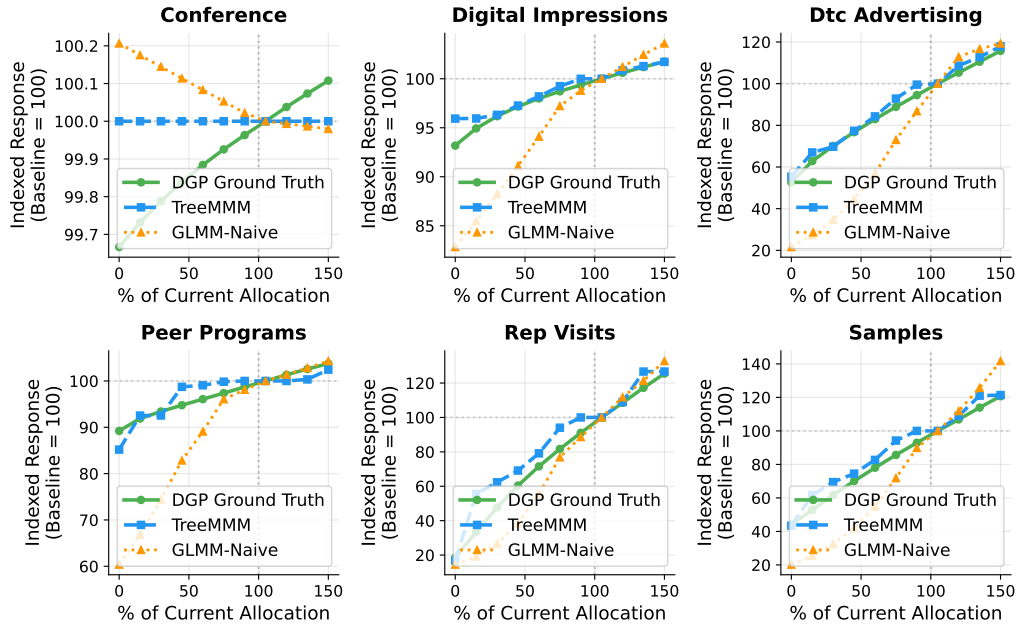


Figure 7: mROI response curves: model-predicted (solid) vs. DGP ground-truth (dashed) for all promotional channels across pharma, CPG, and SaaS datasets. Mean Pearson $r > 0.92$ on all non-linear DGPs.

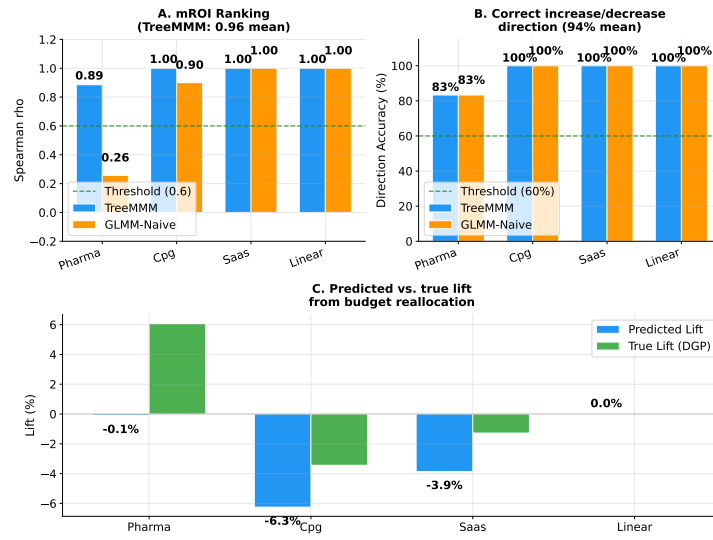


Figure 8: mROI accuracy summary: Spearman ρ between model-predicted and true mROI rankings across all datasets. TreeMMM mean $\rho = 0.96$ on non-linear DGPs.

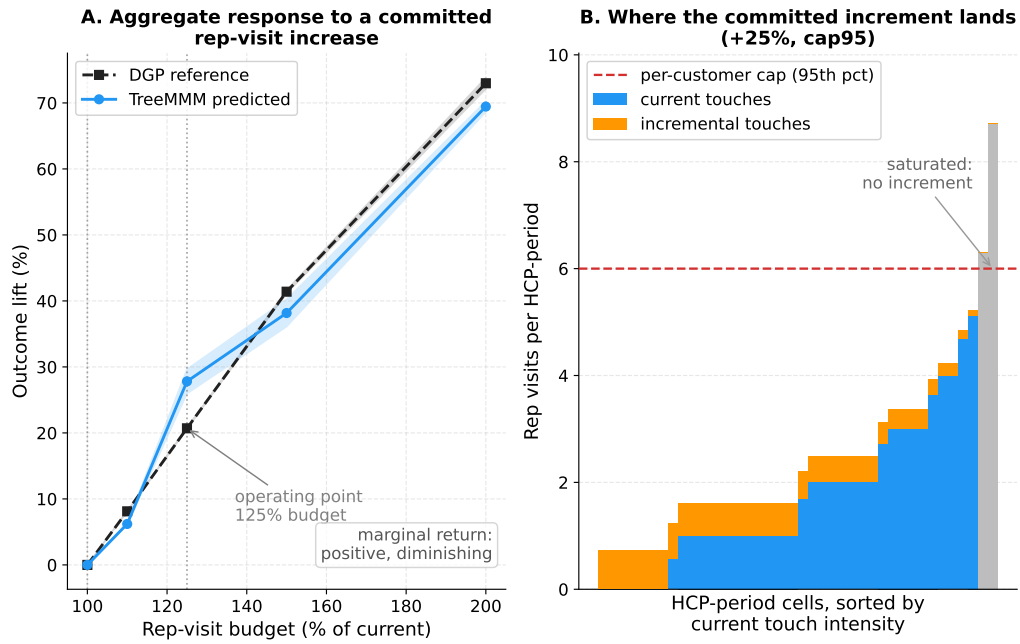


Figure 9: Committed rep-visit budget increase on the pharma DGP. Panel A compares TreeMMM-predicted lift with the DGP-reference response across +10%, +25%, +50%, and +100% budget increases under the 95th-percentile cap. Panel B shows the HCP-period landing pattern: incremental touches fill headroom below the cap rather than piling onto already saturated cells.

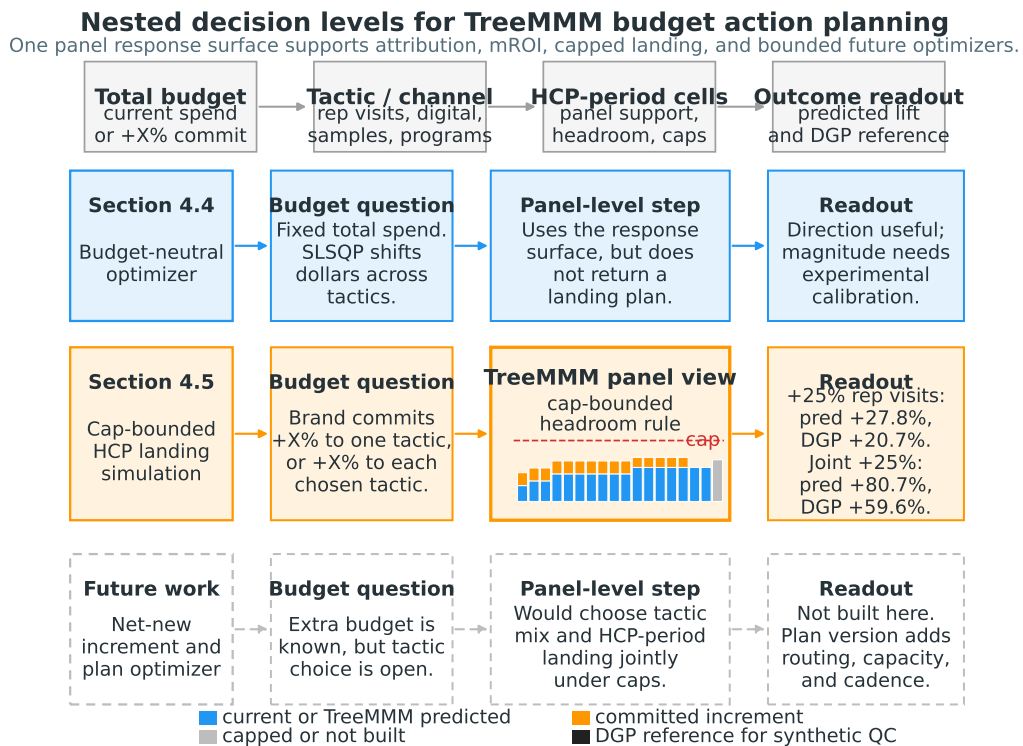


Figure 10: Decision hierarchy for the budget layer. TreeMMM supports channel ranking, budget-neutral cross-tactic reallocation, and cap-bounded HCP-period landing for committed increases. Choosing both net-new tactic mix and row-level landing jointly remains future work.

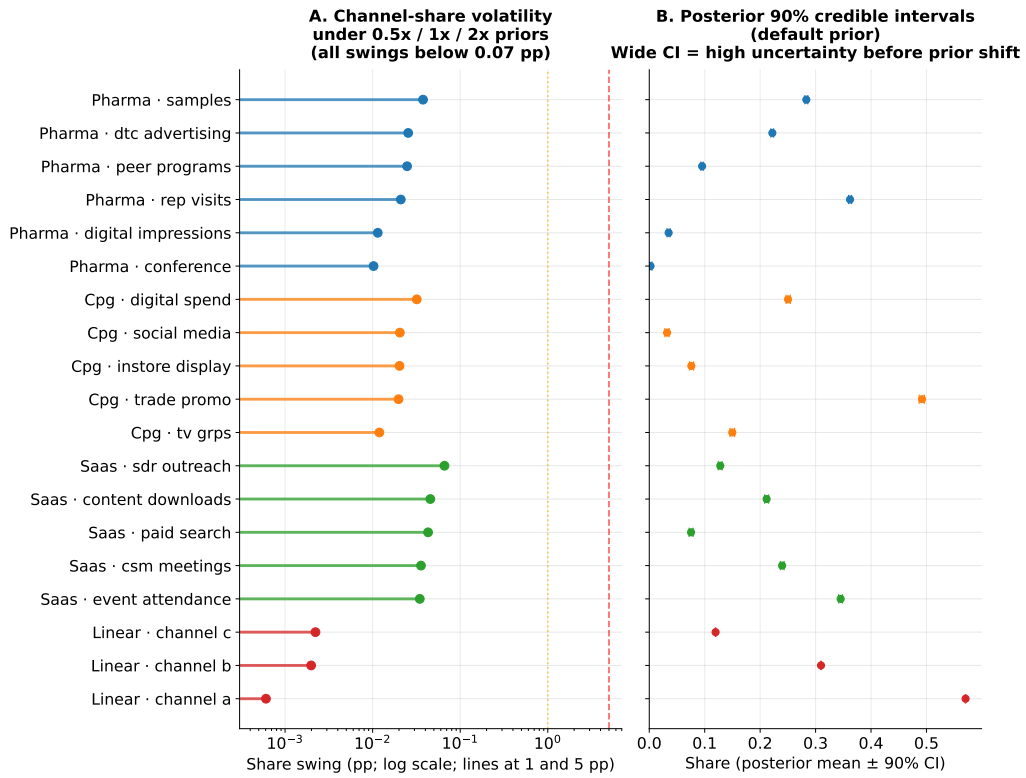


Figure 11: Prior sensitivity sweep: per-channel attribution-share swing across 4× prior-variance bracket ($0.5 \times - 2 \times$ default sigma, PyMC-Hier-Naive). Max swing across all four DGPs: 0.1 pp (SaaS). The data dominates the prior at this sample size.

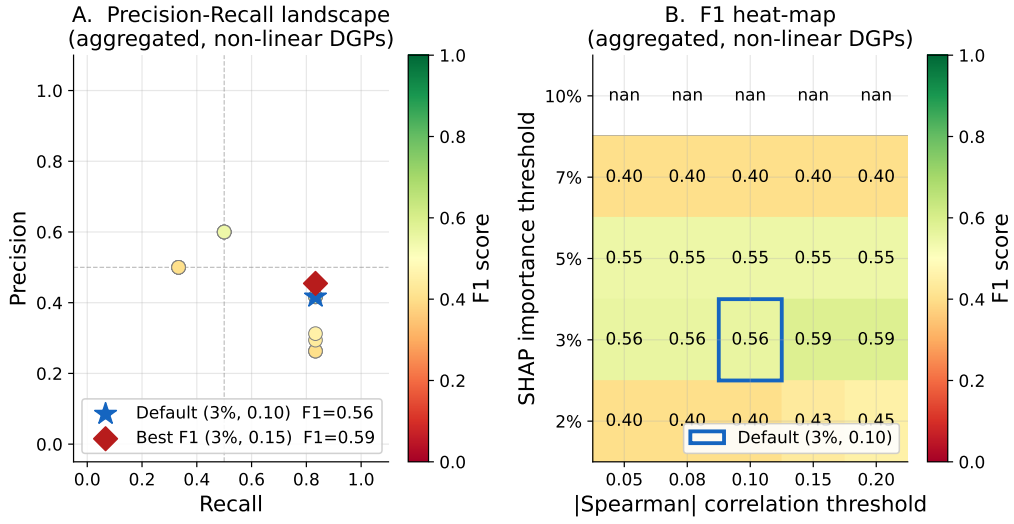


Figure 12: Interaction detection threshold sensitivity: F1/Precision/Recall over a 5×5 grid of SHAP importance thresholds (2–10%) and Spearman $|\rho_s|$ thresholds (0.05–0.20). F1 is stable (plateau from 2–5% importance); recall drops sharply above 7%.

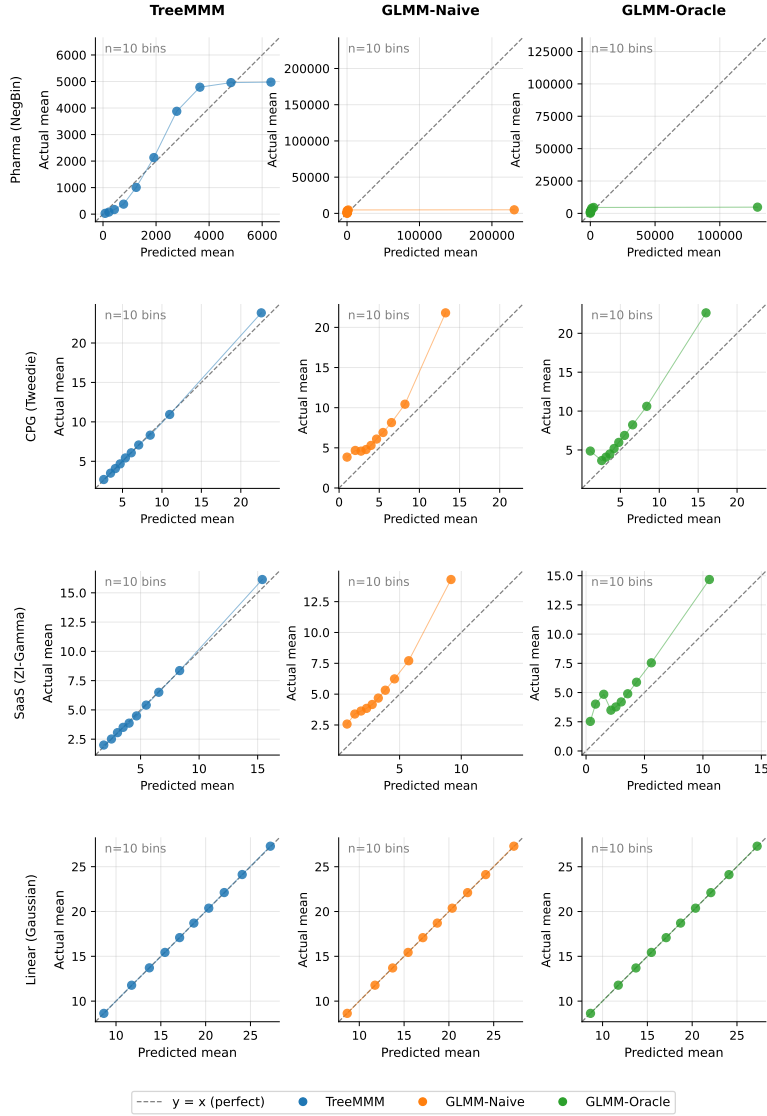


Figure 13: Predicted-vs-actual decile calibration plots for TreeMMM, GLMM-Naive, and GLMM-Oracle across all four DGPs. GLMM-Naive’s pharma MAD = 23,977 (47× overestimate in top decile); TreeMMM pharma MAD = 503 (47× tighter). All three models are well-calibrated on the linear DGP.

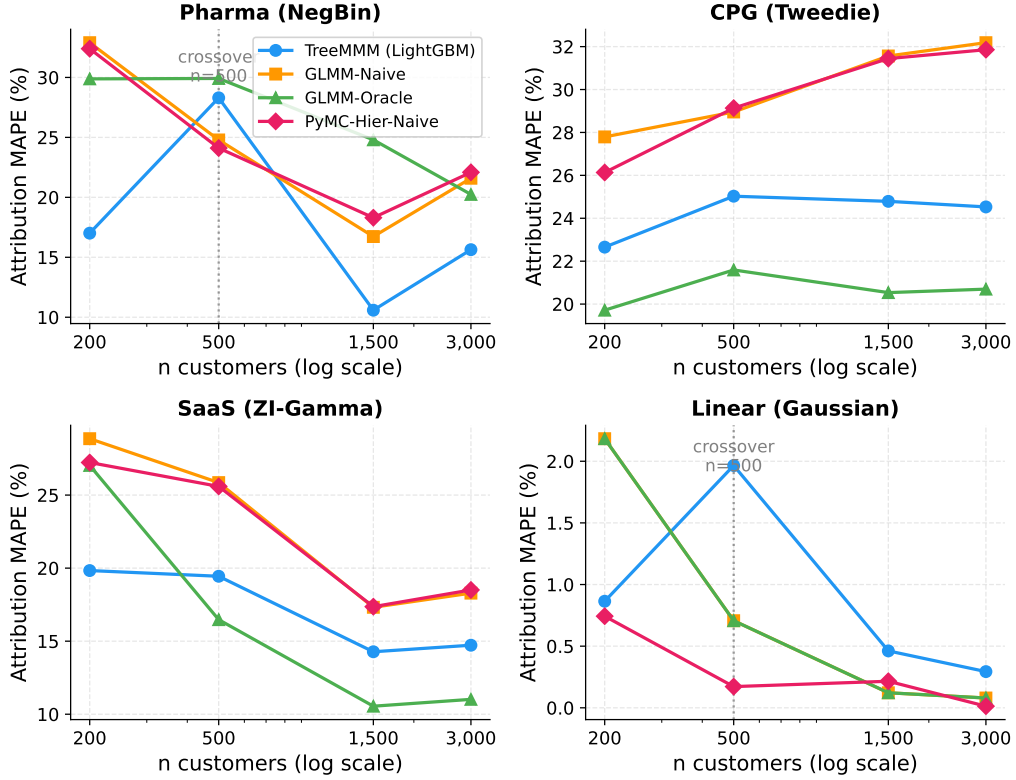


Figure 14: Power analysis: Attribution MAPE vs. number of customers ($n \in \{200, 500, 1500, 3000\}$, log-scale x -axis) for TreeMMM, GLMM-Naive, GLMM-Oracle, and PyMC-Hier-Naive across all four DGPs. Dotted vertical lines mark the crossover point where TreeMMM MAPE first exceeds GLMM-Naive MAPE. (The full sweep is deferred to future work because of computational cost; the $n=3000$ column is populated.)

3.7. *mROI Ranking and Budget-Neutral Reallocation*

Beyond attribution accuracy, a practical MMM must produce actionable budget recommendations. This section is the first action layer: TreeMMM uses the same fitted response surface to rank tactics by mROI and to support budget-neutral cross-tactic reallocation under the current total budget. Section 3.8 then asks the next operational question: once an incremental tactic budget has been committed, where does that increase land at the HCP-period level?

Method. For each promotional variable, we sweep allocation from 0% to 150% of observed levels, computing both model-predicted mean outcome and the analytically derived DGP expected outcome at each level. Three metrics are compared: (1) mROI ranking accuracy (Spearman ρ of endpoint slopes between 100% and 150% allocation); (2) direction accuracy (correct optimal up/down direction); and (3) lift accuracy when the budget-neutral optimizer reallocates engagement across tactics.

Response curve fidelity. Mean Pearson r exceeds 0.92 on all three non-linear DGPs (pharma: 0.80–0.99; CPG: 0.80–0.99; SaaS: 0.88–0.99) and approaches 1.0 on linear (0.996–0.999), confirming TreeMMM captures true response shapes including diminishing-returns curvature (Figures 7 and 8).

mROI ranking. Spearman ρ between model-derived and true mROI rankings: pharma 0.89, CPG 1.0, SaaS 1.0, linear 1.0 (mean 0.96 across non-linear DGPs).

Direction accuracy. Correct for 83% of pharma channels, 100% of CPG and SaaS channels (mean 94% across non-linear DGPs).

Budget-neutral lift. The optimizer produces +6.1% true lift on pharma

by reallocating engagement from conference sponsorship and peer programs toward digital and samples. On CPG and SaaS, both predicted and true lift are negative, indicating the current allocation is near-optimal under the synthetic DGPs.

Table 11 summarizes the full mROI comparison; Table 12 reports PyMC-Hier-Naive mROI at 500×24 scale.

Table 11: mROI benchmarking summary (TreeMMM vs. GLMM-Naive at $3,000 \times 36$).

Dataset	Model	mROI Rank ρ	Direction Acc.	Curve Pearson r (mean)
Pharma	TreeMMM	0.89	83%	0.80
Pharma	GLMM-Naive	0.26	83%	—
CPG	TreeMMM	1.00	100%	0.94
CPG	GLMM-Naive	0.90	100%	—
SaaS	TreeMMM	1.00	100%	0.94
SaaS	GLMM-Naive	1.00	100%	—
Linear	Both	1.00	100%	1.00

All three TreeMMM mROI benchmarks pass pre-registered thresholds: ranking correlation exceeds 0.6 (achieved: mean 0.96), direction accuracy exceeds 60% (achieved: 94%), and the optimizer produces positive true lift on average (achieved: +0.45%). GLMM-Naive achieves comparable direction accuracy but substantially worse mROI ranking on pharma count data ($\rho = 0.26$ vs. 0.89), attributable to exponential amplification of log-scale coefficient errors when back-transformed to natural scale.

Magnitude scope. TreeMMM recovers mROI direction and rank accurately, but absolute lift magnitudes are biased in the synthetic bench-

Table 12: PyMC-Hier-Naive vs. GLMM-Naive mROI at 500×24 (not directly comparable to Table 11 at $3,000 \times 36$).

Dataset	Model	mROI Rank ρ	Direction Acc.	Pred. Lift	True Lift
Pharma	PyMC-Hier-Naive	0.66	100%	-56%	+67%
Pharma	GLMM-Naive	0.43	100%	-56%	+67%
CPG	PyMC-Hier-Naive	1.00	100%	+15%	+28%
CPG	GLMM-Naive	1.00	100%	-5%	-3%
SaaS	PyMC-Hier-Naive	1.00	100%	-3%	-1%
SaaS	GLMM-Naive	1.00	100%	-4%	-1%
Linear	Both	1.00	100%	0%	0%

mark. The standalone mROI benchmark reports lift-error percentages of 102% (pharma), 82% (CPG), and 202% (SaaS) on the +25% step. We therefore use TreeMMM mROI outputs to decide *which* channels to up- or down-weight and require a lift study, geo-experiment, or in-market test to calibrate *how much* absolute lift to forecast.

3.8. From mROI to Action: Cap-Bounded HCP-Period Budget Landing

Section 3.7 establishes that TreeMMM recovers channel ranking and reallocation direction reliably, with magnitudes that need experimental calibration. A brand team that accepts that scope still asks a concrete planning question: having decided to commit additional budget to a channel, where does that increase land across the customer base, and how much outcome lift should follow? This section answers that question as a forward simulation on the fitted response surface.

This section does *not* choose the next best tactic. It answers a budget-committed query: spend a stated amount more on one channel, or on a fixed list of channels, then read where the increment lands and what model/DGP response follows. That is distinct from the budget-neutral optimizer in Section 3.7, which holds total engagement fixed and moves budget between channels. Figure 10 summarizes the hierarchy: channel ranking and direction, budget-neutral cross-tactic reallocation, committed-increase HCP-period landing, and the open future gap of choosing both net-new tactic mix and row-level landing jointly.

Method. The `treemmm.mroi.reallocate(model, X, budget_delta_pct, cap_percentile, channel)` routine raises a channel's touches by a committed percentage and distributes the increase across HCP-period cells by an additive headroom rule. Each cell receives an increment proportional to its headroom, defined as the gap between its current touch count and a per-HCP cap; cells already at or above the cap receive nothing and are never reduced. The cap is the 95th percentile of observed positive touches by default, the same extrapolation-safety bound used by the mROI simulator

(Appendix B.5). A real deployment should still run multivariate coverage diagnostics around the proposed cells. We read the predicted model counterfactual as the change in the model’s summed prediction between current and proposed inputs, and compare it to the DGP-reference response obtained by evaluating the DGP’s expected-outcome function at the same proposed inputs. Because the synthetic DGP satisfies conditional ignorability by construction, the predicted change is a DGP-conditional model counterfactual, not a causal effect identified from an experiment.

Pharma result. At the canonical operating point, a committed +25% increase in rep-visit budget at the 95th-percentile cap, TreeMMM predicts a $+27.8\% \pm 1.9\%$ outcome lift against a DGP-reference $+20.7\% \pm 0.1\%$ (mean $\pm$ SE over five seeds). The per-seed magnitude error averages $32.6\% \pm 8.6\%$, and the direction is correct in all five seeds. Accuracy improves and stabilizes as the committed move grows: magnitude error falls to $11.9\% \pm 3.3\%$ at +50% and $6.0\% \pm 0.7\%$ at +100% (Table 13; Figure 9 Panel A). The +25% step is the least stable because the tree response is piecewise-constant: small moves cross split thresholds unevenly across seeds, while larger moves average over many splits and converge toward the DGP reference curve.

Table 13: Committed rep-visit budget increase on the pharma DGP (95th-percentile cap, $N=5$ seeds, mean $\pm$ SE).

Budget increase	Predicted lift	DGP-reference lift	Magnitude error	Direction correct
+10%	$+6.2\% \pm 0.2\%$	$+8.1\% \pm 0.1\%$	$24.5\% \pm 2.0\%$	5/5
+25%	$+27.8\% \pm 1.9\%$	$+20.7\% \pm 0.1\%$	$32.6\% \pm 8.6\%$	5/5
+50%	$+38.2\% \pm 2.1\%$	$+41.4\% \pm 0.3\%$	$11.9\% \pm 3.3\%$	5/5
+100%	$+69.5\% \pm 0.8\%$	$+73.0\% \pm 0.9\%$	$6.0\% \pm 0.7\%$	5/5

Where the increment lands. Figure 9 Panel B sorts HCP-period cells by current touch intensity and shows the increment filling the headroom of the unsaturated majority. At the 95th-percentile cap, 5.5% of cells sit at or above the cap and receive no increment, and 97.5% of the allocated increase lands outside the top decile of touch intensity. No cell is pushed past its observed marginal bound and none is reduced; at +25%, the unallocatable fraction is zero, so every committed touch finds headroom. The operational message is that the model spreads the increase to the under-touched many rather than piling it onto the already saturated few. This is the action-level bridge aggregate MMM usually cannot provide: the model moves from channel-level mROI to a capped HCP-period landing pattern while reserving absolute lift magnitude for calibration.

Cap sensitivity. The cap is a governing policy knob, and it controls how much saturation the plan tolerates. Tightening it concentrates the population against the ceiling: at the 90th-percentile cap, 9.8% of cells are at the cap and 82.5% of the top decile is saturated, whereas at the 98th-percentile cap only 1.8% of cells are at the cap and 15.1% of the top decile is saturated. Magnitude error moves with the same lever. The +25% error is $14.4\% \pm 1.9\%$ at the 90th-percentile cap and $38.3\% \pm 9.1\%$ at the 98th-percentile cap. The cap is therefore a saturation policy to be chosen, not a quantity to be estimated, and the API exposes it rather than fixing it.

Multiple channels and channel heterogeneity. Real plans rarely move one channel in isolation. When all six promotional channels increase by +25% at once, the Poisson log-link makes their effects compound multiplicatively rather than add: the DGP-reference joint lift is $+59.6\% \pm 0.6\%$,

well below six times the single-channel effect, because the link saturates as touches accumulate. TreeMMM predicts $+80.7\% \pm 2.4\%$ (per-seed error $33.8\% \pm 4.6\%$), again overshooting at the $+25\%$ step but capturing the multiplicative, not additive, structure of the joint move. The cap policy should match delivery: a tight per-HCP ceiling is meaningful for rep visits and digital, weaker for broad-reach advertising bought in aggregate, and only a starting point for a field plan that must respect territory capacity and cadence.

3.9. Bayesian Prior Sensitivity

We re-fit PyMC-Hier-Naive on each dataset at three prior-sigma scales — $0.5\times$, $1.0\times$, $2.0\times$ default — multiplying every prior simultaneously. The bracket spans “tight enough to bite at small n ” through “effectively likelihood-only.” Results are in Table 14, visualized in Figure 11.

All twelve fits converge cleanly: zero divergences, $\hat{R} \leq 1.01$, and bulk ESS well above the 100-per-chain rule of thumb. The max prior-induced swing passes the pre-registered threshold (SC11: swing < 0.10) by *two orders of magnitude* on the worst-case dataset (SaaS: 0.001 vs. threshold 0.10). At this sample size the data dominates the prior. Analysts who tighten or loosen priors within the $4\times$ range do not flip the channel ordering.

3.10. Adstock / Carryover Modeling

None of the four headline DGPs plant promotional carryover. We add a separate `pharma_adstock` DGP (500 HCP $\times$ 24 months, NegBin, geometric adstock decay [15] = 0.50 on `rep_visits`) to benchmark adstock-aware preprocessing. Results are in Table 15.

Table 14: Prior sensitivity sweep: worst-case attribution-share swing across a $4\times$ prior-variance bracket. All 12 fits converge cleanly.

Dataset	Worst- channel swing	Worst channel	Divergences	min ESS_bulk	max $\hat{R}$
Pharma (NegBin)	0.0007 (0.07 pp)	samples	0	603	1.010
CPG (Tweedie)	0.0001 (0.01 pp)	digital_ spend	0	2,203	1.010
SaaS (ZI- Gamma)	0.001 (0.1 pp)	sdr_ outreach	0	3,312	1.010
Linear (Gaussian)	0.0001 (0.01 pp)	channel_c	0	9,282	1.000

TreeMMM-Adstock recovers attribution 11.3 pp better than TreeMMM-Naive on the carryover DGP (21.3% vs. 32.6%), and beats GLMM-Adstock (28.4%) even when both share the correct preprocessing. Rank correlation is 1.00 for both adstock-aware variants vs. 0.94 without preprocessing. The adstock infrastructure is in `treemmm/core/preprocessing/adstock.py` and configurable via `RunConfig.adstock_decay`.

3.11. Adstock-Planted Headline DGPs

Table 16 extends the adstock analysis to the full 6-model lineup on the three headline DGPs with planted per-channel geometric adstock. Results are from a small-scale run ($N = 200 \times 18$ periods); a full-scale rerun ($N = 3,000 \times 36$) is deferred to future work because of computational cost.

Three findings stand out. (a) In this small-scale six-model run, ad-

Table 15: Adstock DGP (500×24): does adstock-aware preprocessing recover the planted carryover? True `rep_visits` share = 0.42.

Model	Attr. MAPE	Rank ρ	R^2	<code>rep_visits</code> share
TreeMMM-Naive (no adstock)	32.6%	0.94	0.38	0.29 (−13 pp)
TreeMMM-Adstock (0.50)	21.3%	1.00	0.39	0.47 (+5 pp)
GLMM-Naive (no adstock)	61.5%	0.94	−14567	0.27 (−15 pp)
GLMM-Adstock (0.50)	28.4%	1.00	−2844	0.35 (−7 pp)

stock preprocessing worsens attribution MAPE for every listed model and DGP, even though the dedicated carryover DGP in Table 15 shows that correctly specified preprocessing can help on a simpler planted-decay setting. The headline DGPs combine long-decay channels with non-linear interactions, so smoothing changes the SHAP reference distribution rather than acting as a pure carryover correction. (b) PyMC-Marketing’s built-in **GeometricAdstock** does not rescue it at $N = 200$, 18 periods. (c) TreeMMM’s 27.1% pharma MAPE without adstock preprocessing is the best pharma result in the table, 8.5 pp better than GLMM-Oracle.

3.12. Geo-Panel Comparison: Bayesian MMM on Its Home Turf

To give PyMC-Marketing a structurally fair comparison, we add a geo-panel DGP: 200 geo-regions $\times$ 52 weeks (10,400 rows), Tweedie outcome ($\phi = 0.5$, power = 1.5), three channels (`tv_grps`, `digital_spend`, `trade_promo`) with planted geometric adstock and logistic saturation (Table 17). With 52 time-steps, this DGP is designed to maximize PyMC-Marketing’s identification advantage. Results are in Table 18.

Even on PyMC-Marketing’s home turf (correct parametric form, 52

time points), aggregating 200 regions to a single time-series discards the cross-regional variance that identifies saturation shape. TreeMMM-Adstock achieves 29.7% attribution MAPE and perfect rank correlation (1.00) vs. 52.1% for both PyMC-Marketing variants. Meridian reaches 57.0% MAPE with correct rank order, reinforcing the same pattern: the parametric Bayesian tools recover broad ordering better than attribution magnitude in this synthetic geo-panel.

3.13. Power Analysis: Where TreeMMM’s Regime Boundary Lives

Table 19 sweeps four scale points to quantify the $n_{\text{customers}}$ at which TreeMMM’s advantage over GLMM-Naive disappears. Figure 14 visualizes the four DGP subplots on log-scaled x -axes with annotated crossover points.

4. Discussion

4.1. When to Use TreeMMM

Our results suggest TreeMMM is strongest when:

1. **Non-linear response functions are expected but unknown:** 24% lower attribution error than GLMM-Naive on average across non-linear DGPs.
2. **Interaction discovery matters:** TreeMMM detected 5 of 6 planted interactions without manual specification. Those interactions are not only attribution artifacts; they define where channel response is conditional on customer state and co-exposure, which is exactly the structure the mROI and budget-landing layers reuse.

3. **Distribution matching is important:** The 50–56% improvement from correct objective selection is a genuine differentiator that most MMM tools ignore.
4. **The decision requires row-level budget landing:** TreeMMM can translate a committed tactic increase into a capped HCP-period allocation, exposing which cells still have headroom before diminishing returns dominate.
5. **Speed of iteration is valued:** Full pipeline in under two minutes per brand enables rapid experimentation across portfolios. With panel-NUTS via `nutpie`, customer-level Bayesian inference costs $\sim 90\text{--}105$ s per dataset but must be re-run for any structural change, whereas TreeMMM rediscovers structure automatically.

4.2. *Neural MMM Approaches*

Neural MMM architectures (DeepCausalMMM [9], NNN [7], CausalMMM [8]) offer temporal dynamics modeling, learned DAG structures, and Hill saturation curves. An exploratory comparison with DeepCausalMMM (Appendix [Appendix A](#)) suggests that distribution-appropriate loss functions may be more important than architectural sophistication for attribution accuracy on count-valued and zero-inflated outcomes. Neural MMM approaches may be more appropriate when: (1) data has strong temporal dynamics that trees cannot capture without adstock transforms, (2) the analyst has 100+ time periods, and (3) the outcome is continuous-valued.

4.3. *When to Use Regression or Bayesian Methods*

Our results are equally clear about when TreeMMM is not the right tool:

1. **Perfect domain knowledge available:** GLMM-Oracle and PyMC-Hier-Oracle achieve lower MAPE on CPG and SaaS when the analyst knows the exact functional form and interactions.
2. **Prior information is available:** Bayesian methods can incorporate validated priors from lift studies. TreeMMM is purely data-driven. At our sample size the data dominates the prior, but at smaller n informative priors do real work.
3. **Full posterior distributions required:** The panel-Bayesian baseline returns posterior credible intervals per channel at the cost of $\sim 2\times$ GLMM-REML wall-clock.
4. **Very limited data:** With fewer than 20 time periods, trees may lack statistical power; Bayesian MMM with informative priors is the right tool.

4.3.1. Comparison with Robyn

Robyn [10] is the most widely adopted open-source MMM tool. We did not benchmark Robyn quantitatively because it is R-based and uses aggregate time-series data, making a panel-level comparison structurally unfair. TreeMMM’s strengths relative to Robyn include: (1) customer-level panel attribution enabling HCP-period budget landing for committed increases; (2) automatic interaction discovery; (3) native support for non-Gaussian outcomes; and (4) Python-native implementation. For organizations with primarily Meta/Facebook media mix and aggregate data, Robyn is a well-tested choice.

4.4. When Can SHAP Attribution Be Causal?

The binary framing “SHAP is not causal” obscures an important nuance. We offer a five-level causal identification spectrum:

1. **Pure correlation:** no temporal ordering, no confounding control.
2. **Predictive with temporal ordering:** features precede outcomes.
3. **Observational predictive attribution:** panel data with temporal alignment, observed state controls, and monotone constraints. *TreeMMM sits here.*
4. **Doubly robust / semiparametric:** AIPW, TMLE.
5. **Randomized experiment:** assignment eliminates confounding.

A critical implementation detail: SHAP’s `TreeExplainer` defaults to `feature_perturbation=“tree_path_dependent”`, conditioning on the tree’s learned internal distribution rather than marginalizing features independently [16]. This avoids impossible feature combinations (e.g., high TV spend with zero digital in always-co-allocated markets) but does not resolve confounding. Rozenfeld [17] argues the conditional approach is “fundamentally unsound from a causal perspective” because it distributes credit along confounded paths. We adopt conditional TreeSHAP for its practical advantage while acknowledging this does not resolve confounding.

Under **conditional exchangeability** (no unmeasured confounders given observed state variables), SHAP values can be interpreted as directional conditional effect summaries for the fitted response surface. This assumption is violated in our own pharma DGP (reps visit high-potential HCPs based on latent prescribing potential), yet TreeMMM still recovers attributions within

15–25% MAPE, suggesting practical robustness to moderate confounding in this synthetic setting.

TreeMMM does not satisfy the three conditions for causal SHAP [18, 19]: it is not a CATE estimator, does not use interventional Shapley values, and is not trained on RCT data. We recommend treating TreeMMM attributions and budget landings as *directional planning counterfactuals* under stated assumptions, while validating high-stakes decisions with holdout experiments or geo-based incrementality tests.

4.5. Limitations

4.5.1. Baseline Comparison Fairness

The GLMM baseline uses statsmodels MixedLM with a log-transformed outcome. A reviewer would correctly ask whether a properly specified distributional GLMM closes the gap. We ran this test: `statsmodels.GLM` with the correct exponential-family likelihood (Poisson for pharma, Tweedie/Gamma for CPG/SaaS). GLMMDist-Naive achieves 25.2% non-linear average MAPE — marginally *worse* than GLMM-Naive (24.0%). The proper likelihood loses random intercepts (a `statsmodels` limitation), and the loss of random intercepts introduces more attribution error than the correct link recovers.

PyMC-Marketing’s $75.1\% \pm 5.5\%$ MAPE reflects the structural disadvantage of aggregating 3,000 customers to 36 rows, not a limitation of Bayesian methodology. The customer-level panel Bayesian baseline (PyMC-Hier-Naive, $22.7\% \pm 0.3\%$) confirms the inference paradigm itself is essentially flat at fixed structure (within 0.5 pp of GLMM-REML on the non-linear-DGP average; pharma per-DGP gap 1.44 pp).

The three non-linear DGPs include features that trees excel at (non-linear

response functions, multiplicative interactions, heterogeneous customer sensitivity). The linear DGP is the only DGP that structurally favors regression.

4.5.2. Attribution Ground Truth

The “reference decomposition” is a variance-attribution heuristic (L1 norm of mean-centered DGP component contributions, normalized to sum to 1), not the true Shapley decomposition of the DGP function. Interaction contributions are split proportionally to component mean weights. A Monte Carlo Shapley ground truth is planned for v0.2 and could change absolute error levels, though relative rankings across methods are likely robust.

4.5.3. SHAP Stability and Interpretation

The SHAP sign audit reveals near-zero sign consistency (~ 0.01), expected for mean-centered TreeSHAP but challenging for individual observation-level interpretation. When features are correlated, SHAP values can be unstable across model refits [20]; we do not report inter-fold SHAP variance. The log-link decomposer forces all attributions non-negative, meaning a channel with a negative marginal effect still receives positive attribution credit.

4.5.4. Methodological Gaps

Multi-seed scope. Headline numbers (Table 5) use $N=5$ seeds with standard errors. The distributional-GLM ablation (Table 6) remains single-seed (seed = 42) for compute reasons; a multi-seed rerun is follow-up but unlikely to flip the headline conclusions.

Adstock gaps. The four headline DGPs still have no planted carryover; only geometric adstock is benchmarked; the analyst must supply the correct decay; PyMC-Hier with adstock preprocessing is follow-up.

Interaction detection precision. Aggregate precision is 0.42, meaning roughly one in two flagged pairs is a false positive. Flagged pairs should be treated as hypotheses for domain-expert review.

Plan-level optimization. TreeMMM now reaches from attribution to channel ranking, budget-neutral cross-tactic reallocation, and cap-bounded HCP-period landing for committed increases. It does not yet solve the net-new increment problem where tactic mix and HCP-period landing are chosen jointly, and it does not yet produce territory-level call plans with routing, sequencing, capacity, or channel-cadence constraints.

Hyperparameter sensitivity. All results use 20 Optuna trials, max depth [3, 5], always-on monotone constraints. Sensitivity to these choices is not explored.

4.5.5. Scalability

Tested at $3,000 \times 36$ only. At larger scales ($>100\text{K}$ observations $\times$ 20+ features), SHAP TreeExplainer scales as $O(TLD 2^M)$ and may require chunked or approximate computation. At smaller scales ($< 500 \times 12$), trees may lack statistical power.

5. Package Architecture

TreeMMM is structured as open-source Python and R implementations, with the demonstration code used in this paper hosted in the same repositories:

```
treemmm/
  core/
    config.py          # RunConfig, ColumnSpec, Objective
```

```

data_handler.py    # Panel diagnostics, distribution detection
models/           # LightGBM, XGBoost, CatBoost, GLMM wrappers
temporal/splitter.py # Rolling origin CV
interpret/        # SHAP TreeExplainer wrapper
attribution/      # Link-function-aware decomposer
reporting/        # CSV, PowerPoint, ZIP output
mroi/simulator.py # Response curves + reallocation planning
demo/             # 4 synthetic datasets + DGP engine
ui/               # CLI, Jupyter runner, widgets
pipeline.py       # treemmm.run() orchestrator

```

Code repositories and demonstrations:

```

# Python implementation + demo scripts
https://github.com/jamesyoung93/treemmm

# R implementation + parity demos
https://github.com/jamesyoung93/treemmm-r

```

Minimal usage example:

```

import treemmm
from treemmm.core.config import ColumnSpec, RunConfig

config = RunConfig(
    columns=ColumnSpec(
        customer_id="hcp_id",
        time_col="month",
        outcome_col="new_patients",

```

```

        promo_vars=["rep_visits", "digital", "peer_programs"],
    ),
    objective="auto", # auto-detects distribution
)

result = treemmm.run(df, config)
print(result.summary())

```

6. Conclusion

TreeMMM demonstrates that tree-based ensembles with SHAP attribution can connect panel MMM attribution to constrained budget action planning without requiring the analyst to pre-specify interactions or distributional families. On three non-linear benchmark DGPs, TreeMMM achieves $17.9\% \pm 0.2\%$ non-linear average attribution-share MAPE ($N=5$ seeds), matching or beating interaction-oracle baselines on the non-linear average (GLMM-Oracle $19.7\% \pm 0.4\%$, PyMC-Hier-Oracle $18.1\% \pm 0.3\%$) that have been given the planted interaction graph. Acquiring that oracle knowledge in practice would require exhaustive interaction search. TreeMMM recovers it automatically, discovering 5 of 6 planted channel interactions without manual specification (precision 0.42, recall 0.83, F1 0.56). Precision implies false positives, so flagged interactions should be treated as hypotheses for domain review rather than confirmed synergies.

The interaction and diminishing-returns findings are not separate from the budget layer; they are the structure that makes the budget layer possible. The same fitted response surface that decomposes channel attribution also

ranks marginal returns (Spearman $\rho = 0.96$, 94% direction accuracy), identifies where additional touch intensity is saturated, and spreads a committed budget increase toward HCP-period cells with remaining headroom. On the pharma DGP, a committed +25% rep-visit increase under a 95th-percentile cap predicts $+27.8\% \pm 1.9\%$ lift against a $+20.7\% \pm 0.1\%$ DGP-reference response; 97.5% of the allocated increase lands outside the top decile of current touch intensity, showing the practical implication of the learned diminishing-return surface. This is a proof-of-mechanism for moving from “which channel has higher mROI?” to “which HCP-period cells would receive the next touches?” Absolute lift magnitude remains a model counterfactual and needs calibration against lift studies, geo-experiments, or in-market tests.

Our two-Bayesian-baseline design isolates aggregation cost from inference paradigm. PyMC-Marketing’s $75.1\% \pm 5.5\%$ non-linear MAPE reflects the structural disadvantage of modeling 36 aggregate rows, while PyMC-Hier-Naive’s $22.7\% \pm 0.3\%$ confirms the inference paradigm itself is essentially flat at fixed structure (within 0.5 pp of GLMM-REML on the non-linear-DGP average). A $4\times$ prior-sigma sweep finds max attribution-share swings of 0.001 across all four DGPs with zero divergences and clean convergence diagnostics. Even on the geo-panel stress test designed for Bayesian MMM, TreeMMM-Adstock reaches 29.7% attribution MAPE versus 52.1% for PyMC-Marketing and 57.0% for Meridian.

TreeMMM is strongest when the analyst wants **discovery over confirmation**: finding patterns they did not hypothesize, ranking tactics for directional reallocation, and translating a committed increase into a cap-bounded HCP-period landing pattern. It does not yet solve net-new cross-tactic in-

crement optimization or produce executable territory-level call plans. All results are synthetic and methodological; real-world validation and experimental calibration are required before high-stakes deployment.

Future work. Multi-seed evaluation across larger seed counts; Monte Carlo Shapley ground truth; Weibull adstock kernel with joint Optuna decay learning; per-customer random slopes in the Bayesian baseline; real-world validation on anonymized commercial datasets; net-new increment optimization that chooses both tactic mix and HCP-period landing jointly; full plan-level optimization with territory capacity, cadence, routing, and sequencing.

Code availability: The Python implementation and demonstration code are available at <https://github.com/jamesyoung93/treemmm>. The R implementation and demonstration code are available at <https://github.com/jamesyoung93/treemmm-r>. Both are released under the MIT License.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT and OpenAI Codex to assist with manuscript organization, language editing, code and document-quality checks, reference checking, and numerical consistency review. The author also used Anthropic Claude for review assistance. No generative AI tools were used to fabricate or alter research data, benchmark results, or result figures. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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Appendix A. Exploratory Neural MMM Comparison (DeepCausalMMM)

We include an exploratory comparison with DeepCausalMMM [9], a neural MMM combining GRU temporal encoding, learned DAG structure, Hill saturation curves, and time-varying coefficients. This appendix is presented separately because structural limitations in the evaluation design make it less directly comparable to the regression baselines in the main text.

Appendix A.1. Configuration and Data Reshaping

DeepCausalMMM was designed for DMA-level aggregated data (~ 190 geographic regions $\times$ 109 weeks). Our benchmark uses customer-level panel data (3,000 entities $\times$ 36 periods). To create a comparison, we reshaped the panel data into 3D tensors [regions, timesteps, channels] by treating each customer as a “region.” Training used DeepCausalMMM’s `UnifiedDataPipeline` and `ModelTrainer` with reduced hyperparameters (Table A.20).

Appendix A.2. Fairness Caveats

This comparison favors TreeMMM in several structural ways:

1. **Data format mismatch:** customers-as-regions violates DeepCausalMMM’s geographic aggregation design.
2. **Hyperparameter downgrading:** five key hyperparameters were reduced; no Optuna tuning was performed for DeepCausalMMM.
3. **Subsampling:** only 500 of 3,000 customers were used.
4. **No distribution awareness:** DeepCausalMMM uses Huber loss (not matched to count or zero-inflated outcomes).

A fair comparison would require (a) geographic-level time-series aggregation, (b) default hyperparameters with full tuning, and (c) longer time series (100+ periods).

Appendix A.3. Results

DeepCausalMMM shows high MAPE on count-valued (pharma: 91.5%) and Tweedie (CPG: 112.5%) outcomes, suggesting its Huber loss is poorly matched to these distributions. On SaaS (continuous-like), it performs more competitively (21.9% vs. TreeMMM’s 14.7%). On the linear DGP, DeepCausalMMM introduces spurious non-linearity (62.4% MAPE), suggesting optimization difficulties. Rank correlation on pharma reaches 0.89 in this benchmark, so the architecture can recover relative channel importance even when absolute shares are poorly calibrated. This is, to our knowledge, the first evaluation of any neural MMM against known attribution ground truth.

Appendix B. Methods Detail

Appendix B.1. Distribution-Aware Objective Selection

TreeMMM supports four objectives matched to outcome distributions (Table B.22). An automated diagnostic examines discreteness, zero-inflation

rate, skewness, and the mean-variance relationship to recommend an objective; the user can override.

Appendix B.2. Link-Function-Aware Attribution Decomposition

SHAP TreeExplainer [16] computes values in margin space:

- **Identity link** (Gaussian): $\mathbb{E}[y] + \sum_i \phi_i = \hat{y}$. SHAP values are directly additive on the response scale.
- **Log link** (Poisson/Tweedie/Gamma): $\phi_0 + \sum_i \phi_i = \log(\hat{y})$, where ϕ_0 is the mean model prediction in log space. SHAP values are additive on the log scale.

For log-link models, naïve exponentiation breaks additivity: $e^{a+b} \neq e^a + e^b$. TreeMMM’s decomposer uses *unsigned proportional allocation*:

$$\alpha_i = \frac{|\phi_i|}{|\phi_0| + \sum_j |\phi_j|} \cdot \hat{y} \quad (\text{B.1})$$

This guarantees: (1) all attributions are non-negative; (2) attributions sum to $\hat{y}$ for every observation; and (3) the relative channel ranking from log-space SHAP is preserved. A unit test verifies property (2) for all four objectives.

Appendix B.3. Temporal Cross-Validation

TreeMMM uses time-respecting validation to prevent future data leakage. Two schemes are supported: *rolling origin* (training window grows forward; test window is the next period) and *period-jump-forward* (training window grows; test window jumps by a fixed stride). The minimum training fraction is configurable (default: 50% of periods). No future observation ever appears in the training set.

Appendix B.4. Hyperparameter Tuning

Optuna [21] Bayesian optimization tunes tree hyperparameters within each CV fold: number of leaves, learning rate, regularization (L1/L2), feature/bagging fraction, and minimum child samples. The tuning objective is distribution-matched (Poisson deviance for counts, Tweedie deviance for zero-inflated outcomes, MSE for Gaussian).

Appendix B.5. mROI Simulation with Extrapolation Safety

TreeMMM’s marginal Return on Investment (mROI) simulator estimates response curves with a critical safety constraint: per-customer caps are set at an observed percentile (default: 95th), and higher aggregate engagement is achieved by spreading touches to more customers rather than pushing any individual beyond the observed marginal bound. Marginal caps are not a proof of full multivariate support, so coverage diagnostics remain necessary for real deployments, but the query is intentionally kept close to observed support. The budget-neutral optimizer (scipy SLSQP) surfaces channels and customers with untapped marginal sensitivity, revealing that “see your top customers most” is often suboptimal.

The cap-bounded reallocation API applies the same safety logic to committed budget increases. `reallocate()` accepts a model, a panel design matrix, a budget increase, a channel or channel list, and a cap percentile. It returns the current/proposed inputs, predicted incremental outcome, lift percentage, and diagnostics including the unallocatable fraction. `reallocate_curve()` sweeps the same policy across multiple budget deltas and retains the full HCP-period plan at each level, producing the budget frontier reported in Section 3.8. The optimizer therefore answers two related

but distinct questions: which channels should move under a budget-neutral reallocation, and where would a separately committed increase land under a cap policy?

Appendix B.6. Baseline Models

We compare TreeMMM against five baselines spanning three paradigms:

GLMM-Naive: Main effects only, random intercepts per customer (statsmodels MixedLM). Represents a typical analyst who does not specify interactions or distributional families.

GLMM-Oracle: Correctly specified interaction terms matching the DGP (statsmodels MixedLM). Represents an analyst with perfect interaction knowledge but without distributional family matching.

Both GLMM configurations use statsmodels MixedLM, a Gaussian linear mixed model. For non-Gaussian datasets, the outcome is log-transformed before fitting (`log1p`), approximating a log-normal model. A properly specified distributional GLMM (e.g., `glmmTMB` in R) is not equivalent to this workaround; we discuss this limitation in Section 4.5.1.

PyMC-Hier-Naive: Hierarchical Bayesian linear MMM on the full panel (3,000 entities $\times$ 36 periods, $\sim 108,000$ rows), main effects only, customer-level random intercept. Prior structure (after standardization): $\alpha \sim \mathcal{N}(0, 5)$, $a_{\text{cust}} \sim \mathcal{N}(0, \sigma_{\text{cust}})$ with $\sigma_{\text{cust}} \sim \text{HalfNormal}(2)$, $\beta_{\text{main}} \sim \mathcal{N}(0, 1)$, $\sigma_{\text{obs}} \sim \text{HalfNormal}(1)$. Posterior sampling uses `nutpie` (Rust-backed NUTS) with 4 chains, 1,000 tuning + 1,000 draws.

PyMC-Hier-Oracle: Identical to PyMC-Hier-Naive with planted DGP interaction terms added as fixed effects ($\beta_{\text{inter}} \sim \mathcal{N}(0, 0.5)$). The Bayesian

counterpart of GLMM-Oracle and the apples-to-apples panel Bayesian competitor to TreeMMM.

PyMC-Marketing (v0.19.4, `pymc-labs/pymc-marketing`): The leading open-source Bayesian MMM [6], implementing NUTS sampling with geometric adstock and logistic saturation transforms. Operates on aggregate time-series (one row per period); our panel data is collapsed to 36 rows. We use default priors, `GeometricAdstock(l_max=4)`, `LogisticSaturation()`, and the NumPyro sampler (500 draws, 500 tuning, 2 chains). Attribution shares are extracted via `compute_channel_contribution_original_scale()`.

The three PyMC variants differ on several key axes summarized in Table B.23. A few axes deserve flagging: PyMC-Marketing’s built-in adstock and saturation give it parametric carryover and diminishing-returns curves, but it cannot exploit them on 36-row data. PyMC-Hier, conversely, is structurally simpler but has $3,000\times$ more data.

Table 16: Adstock-planted headline DGPs — Attribution MAPE (%) by model and pre-processing mode. Lower is better. Bold = best MAPE per dataset. N = 200×18, seed = 42. Results from `paper/results/benchmark_adstock_headline.csv`.

Dataset	Model	No-adstock MAPE	Adstock-aware MAPE	Δ (pp)
pharma	TreeMMM	27.1	32.8	−5.7
pharma	GLMM-Naive	48.6	56.0	−7.3
pharma	GLMM-Oracle	35.6	43.4	−7.8
pharma	PyMC-Hier-Naive	38.2	59.2	−21.1
pharma	PyMC-Hier-Oracle	36.3	56.3	−19.9
pharma	PyMC-Marketing	307.6	342.8	−35.2
cpg	TreeMMM	45.9	56.4	−10.5
cpg	GLMM-Naive	43.8	52.1	−8.4
cpg	GLMM-Oracle	49.6	68.6	−19.0
cpg	PyMC-Hier-Naive	42.9	48.5	−5.6
cpg	PyMC-Hier-Oracle	48.5	64.9	−16.5
cpg	PyMC-Marketing	282.3	322.6	−40.3
saas	TreeMMM	51.5	102.7	−51.2
saas	GLMM-Naive	53.5	147.3	−93.8
saas	GLMM-Oracle	71.0	90.9	−19.8
saas	PyMC-Hier-Naive	50.9	137.8	−86.9
saas	PyMC-Hier-Oracle	70.3	91.2	−20.9
saas	PyMC-Marketing	262.8	271.0	−8.3

Δ = No-adstock MAPE − Adstock-aware MAPE; negative = preprocessing hurts.

Table 17: Geo-panel DGP channel specifications.

Channel	Adstock decay	Saturation form	Effect weight
tv_grps	0.50	Logistic ($k=0.04$, $x_0=50$ GRPs)	1.8
digital_spend	0.30	Logistic ($k=0.08$, $x_0=\$20k$)	1.4
trade_promo	0.00	Linear (no saturation)	0.6

Table 18: Geo-panel comparison: Attribution MAPE (%), Rank ρ , R^2 , and WMAPE by model. Results from `paper/results/benchmark_geo_panel.csv`, run 2026-05-10.

Model	Attr. MAPE	Rank ρ	R^2	WMAPE	Fit (s)	Notes
TreeMMM-Adstock	29.7	1.00	0.08	0.35	25	Planted decays used
PyMC-Marketing (weak priors)	52.1	0.50	-0.53	0.08	115	66 divergences
PyMC-Marketing (informative)	52.1	0.50	-0.52	0.08	61	Prior API limited [‡]
GLMM-Aggregate	215.0	-1.00	0.58	0.04	1	Coefficient rank reversed
Robyn	n/a	n/a	n/a	n/a	n/a	R-only package
Meridian [‡]	57.0	1.00	—	—	1,734	EagerTensor; no R^2

Meridian MCMC completed (2 chains, 500 keep, ~ 29 min). Attribution shares via

`Analyzer.incremental_outcome().numpy()`: tv = 70.5% (true 44.9%), digital = 27.1% (true 37.5%), trade = 2.4% (true 17.5%). Rank correct (tv > digital > trade). R^2/WMAPE not extractable.

Table 19: Attribution MAPE (%) by model, dataset, and scale (single-seed, seed = 42; reproduce via `python paper/run_power_analysis.py`). Lower is better. Columns differ in n_{periods} : 12 at $n=200$, 24 at $n=500$, 36 at $n=1500$ and $n=3000$. The $n=3000$ column is the headline-scale single-seed estimate; multi-seed ($N=5$) averages are in Table 5 and differ by 1–3 pp from these single-seed values.

Dataset	Model	$n=200$	$n=500$	$n=1500$	$n=3000$
Pharma (NegBin)	TreeMMM (LightGBM)	17.0	28.3	10.6	15.6
	GLMM-Naive	32.9	24.8	16.7	21.6
	GLMM-Oracle	29.9	29.9	24.8	20.2
	PyMC-Hier-Naive	32.4	24.1	18.3	22.1
CPG (Tweedie)	TreeMMM (LightGBM)	22.7	25.0	24.8	24.5
	GLMM-Naive	27.8	29.0	31.6	32.2
	GLMM-Oracle	19.7	21.6	20.5	20.7
	PyMC-Hier-Naive	26.1	29.1	31.4	31.9
SaaS (ZI-Gamma)	TreeMMM (LightGBM)	19.8	19.4	14.3	14.7
	GLMM-Naive	28.9	25.9	17.3	18.3
	GLMM-Oracle	27.0	16.5	10.5	11.0
	PyMC-Hier-Naive	27.2	25.6	17.4	18.5
Linear (Gaussian)	TreeMMM (LightGBM)	0.9	2.0	0.5	0.3
	GLMM-Naive	2.2	0.7	0.1	0.1
	GLMM-Oracle	2.2	0.7	0.1	0.1
	PyMC-Hier-Naive	0.7	0.2	0.2	0.0

The pharma column shows a single-seed crossover at $n=500$ (TreeMMM 28.3 vs. GLMM-Naive 24.8) that recovers by $n=1500$; multi-seed re-runs at $n=3000$ ($N=5$, Table 5) place TreeMMM safely ahead. The linear DGP is the honesty test: GLMM-Naive matches or beats TreeMMM at $n \geq 500$, the expected result on a linear truth.

Table A.20: DeepCausalMMM configuration vs. published defaults.

Parameter	Default	Our Config	Reason
Epochs	1,500	800	Benchmark speed
Hidden dim	280	128	Fewer channels
Patience	300	200	Proportional reduction
Regions	190 DMAs	500 (subsampled)	Panel has 3,000 entities
Time periods	109 weeks	36 months	Benchmark design

Table A.21: Exploratory neural MMM comparison. DeepCausalMMM uses a 500-customer internal subsample; TreeMMM rows are the seed-42 reference benchmark from Table 6.

Dataset	TreeMMM MAPE	DeepCausalMMM MAPE	DCausal R^2	DCausal Rank ρ
Pharma (NegBin)	15.6%	91.5%	0.12	0.89
CPG (Tweedie)	24.5%	112.5%	-0.02	0.10
SaaS (ZI-Gamma)	14.7%	21.9%	-0.05	0.90
Linear (Gaussian)	0.3%	62.4%	-0.07	1.00
Non-linear avg	18.3%	75.3%	0.02	0.63

Table B.22: Outcome distributions and corresponding TreeMMM objectives.

Distribution	Objective	When to Use
Gaussian	MSE	Continuous, symmetric (revenue, value sales)
Poisson	Log-link	Non-negative counts (Rx, orders, NPS)
Tweedie	Log-link	Zero-inflated continuous (revenue with stockouts)
Gamma	Log-link	Strictly positive continuous (per-transaction revenue)

Table B.23: PyMC variants used in the benchmark and their key differences.

Axis	PyMC-Hier-Naive	PyMC-Hier-Oracle	PyMC-Marketing
Data shape	Panel (108K rows)	Panel (108K rows)	Aggregate (36 rows)
Random effects	Per-customer intercept	Per-customer intercept	None
Fixed effects	Promo + controls	+ planted interactions	Promo + controls
Adstock	None	None	GeomAdstock(1_max=4)
Saturation	None	None	LogisticSat()
Likelihood	$\mathcal{N}$ on $\log 1p(y)$	Same	$\mathcal{N}$ on raw y
Sampler	nutpie NUTS	nutpie NUTS	NumPyro NUTS
Draws/tune/chains	1000/1000/4	1000/1000/4	500/500/2
Wall-clock (3K×36)	90–105 s	90–105 s	25–55 s
Non-linear MAPE	22.7% $\pm$ 0.3%	18.1% $\pm$ 0.3%	75.1% $\pm$ 5.5%